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Blastocyst-stage versus cleavage-stage embryo transfer in assisted reproductive technology (Review)

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[Intervention Review]

Blastocyst-stage versus cleavage-stage embryo transfer in assisted reproductive technology

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ABSTRACT

Rationale

Blastocyst-stage transfer has increasingly replaced cleavage-stage transfer in IVF (in vitro fertilisation) programmes. It is thought to improve live birth rates by better synchronising embryo development with the endometrium and allowing self-selection of viable embryos. However, it remains uncertain whether extended culture confers a true biological advantage or mainly reflects selection in vitro, and whether benefits per transfer translate into improved cumulative live birth rates across different patient populations and laboratory settings.

Objectives

To assess the effects of blastocyst-stage (day 5 to 6) embryo transfer compared with cleavage-stage (day 2 to 3) embryo transfer on cumulative live birth rate per woman (defined as the occurrence of at least one live birth resulting from the fresh transfer and all subsequent frozen–thawed embryo transfers derived from the same oocyte retrieval), live birth rate per fresh transfer, and cumulative preterm birth outcomes using the same definition.

Search methods

On 1 October 2024, we searched the Cochrane Gynaecology and Fertility Group Specialised Register of controlled trials, CENTRAL, MEDLINE, Embase, PsycINFO, and CINAHL. In addition, we searched two trial registries, reference lists of relevant papers, and contacted experts in the field for any additional trials.

Eligibility criteria

We included randomised controlled trials (RCTs) that involved women undergoing an assisted reproductive technology cycle and compared blastocyst-stage versus cleavage-stage embryo transfers.

Outcomes

Our critical outcomes of interest were cumulative live birth rate from fresh and frozen-thawed cycles, live birth rate per fresh transfer, and cumulative preterm birth rate from fresh and frozen-thawed cycles. Our important outcomes of interest were clinical pregnancy rate per fresh transfer, miscarriage rate per fresh transfer, embryo freezing rate, and failure-to-transfer rate.

Risk of bias

We assessed the risk of bias in the evidence using Cochrane's RoB 2 tool.

Synthesis methods

We synthesised results for each outcome using meta-analysis where possible (with a fixed-effect Mantel-Haenszel model). Where this was not possible due to the nature of the data, we provided a narrative description of the results. We used GRADE to assess the certainty of the evidence.

Included studies

We included 36 parallel-design RCTs (8389 participants). Three studies reported cumulative live birth rate (cLBR); two studies reported cumulative obstetric and perinatal outcomes; and 19 studies reported live birth rate (LBR) per fresh transfer.

Synthesis of results

Blastocyst-stage transfer may increase cLBR within 12 months after oocyte retrieval (OR 1.24, 95% CI 1.04 to 1.47; I^2 = 70%; 3 studies, 2328 women; low certainty). The high heterogeneity appeared to be explained by one large study in which women underwent an average of two or more embryo transfers, with little or no difference found between strategies (OR 1.02, 95% CI 0.81 to 1.28; 1202 women; low certainty). The meta-analysis result indicates that if 62% of women achieve a live birth after fresh or frozen cleavage-stage transfer, between 63% and 70% would do so with blastocyst-stage transfer.

LBR per fresh transfer was probably higher with blastocyst-stage transfer (OR 1.39, 95% CI 1.23 to 1.56; 19 studies, 4787 women; moderate certainty). For fresh transfers, this suggests that if 32% of women achieve a live birth after cleavage-stage transfer, between 37% and 43% would do so after blastocyst-stage transfer.

Cumulative preterm birth rates following fresh and frozen transfers are probably higher with blastocyst-stage transfers (OR 1.72, 95% CI 1.14 to 2.59; 2 studies, 2194 women; moderate certainty). This suggests that if 3.5% of women experience preterm birth after cleavage-stage transfer, between 3.9% and 8.5% will do so after blastocyst-stage transfer.

The clinical pregnancy rate (CPR) per fresh transfer may be higher with blastocyst-stage transfers (OR 1.31, 95% CI 1.20 to 1.44; I^2 = 52%; 36 studies, 8389 women; low certainty). This suggests that if 39% of women achieve clinical pregnancy after fresh cleavage-stage transfer, between 43% and 48% will likely do so with blastocyst-stage transfer.

There may be little to no difference between groups in miscarriage rate (OR 1.12, 95% CI 0.94 to 1.33; I^2 = 0%; 25 studies; 6674 women; low certainty).

The proportion of women with supernumerary embryos available for cryopreservation for use in a later transfer may be lower in the blastocyst-stage transfer group (OR 0.44, 95% CI 0.38 to 0.52; I^2 = 82%; 17 studies; 4620 women; low certainty).

The failure-to-transfer rate may be higher in the blastocyst-stage transfer group (OR 2.69, 95% CI 1.94 to 3.73; I^2 = 35%; 21 studies; 5145 women; low certainty).

The certainty of evidence was low for most outcomes; it was limited by risk of bias, heterogeneity, imprecision, and poor reporting of randomisation methods.

In sensitivity analyses, the findings were similar when we removed data from studies at high risk of bias.

Authors' conclusions

In the first 12 months of follow-up, cumulative live birth rate (cLBR) may be higher after blastocyst-stage embryo transfer than cleavage-stage transfer in women with a good prognosis for pregnancy. However, while some participants may have used all available embryos within this timeframe, others would have had embryos remaining, so cLBR estimates may be different after complete embryo exhaustion. Fresh blastocyst-stage transfer probably increases live birth rates and may improve clinical pregnancy rates compared to cleavage-stage transfer in women with a good prognosis, but may have little to no effect on miscarriage rate. Cumulative preterm birth (< 37 weeks) probably increases with blastocyst-stage transfer. Embryo freezing may be more common and failure-to-transfer less frequent with cleavage-stage embryos. However, it remains unclear whether these differences translate into higher cumulative live birth rates or instead reflect longer treatment trajectories with increased costs and time to pregnancy. None of these findings should be generalised to women with poor prognoses, who were underrepresented in most included studies.

Future research should prioritise the assessment of cumulative live birth rate at extended follow-up, the measurement of miscarriage rates, and the evaluation of obstetric outcomes, particularly in women with poor prognoses.

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Registration

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PLAIN LANGUAGE SUMMARY

When trying to have a baby through assisted conception, is it better to transfer the embryo to the womb on day 3 or day 5?

Key messages

- The chance of having a live-born baby after one IVF (in vitro fertilisation) cycle may slightly improve when the embryo is transferred on day 5 to 6 ('blastocyst' stage) rather than day 3 ('cleavage' stage), in women with a 'good prognosis' (that is, characteristics that make getting pregnant easier). This measure included all transfers from eggs collected in one cycle (fresh and frozen).
- When looking only at results for embryos transferred fresh (placed into the womb a few days after fertilisation), we found that transferring embryos at the blastocyst stage rather than the cleavage stage probably increases the chance of having a baby.
- Transferring embryos at blastocyst stage rather than cleavage stage probably increases the risk of having a baby born preterm (early) at between 32 and 36 weeks as opposed to at full term (40 weeks).
- More research is needed to understand long-term effects, especially in people with a lower chance of IVF success.

What is assisted conception?

Assisted conception refers to medical treatments used to help people achieve a pregnancy when it has not happened naturally. In vitro fertilisation (IVF) is a fertility treatment where eggs are collected from a woman and fertilised with sperm artificially outside the body, and then one or more embryos are put into the woman's womb (i.e. 'transferred'). Embryos can be transferred a few days after fertilisation ('cleavage' stage, day 2 to 3) or a little later ('blastocyst' stage, day 5 to 6).

Transferring blastocyst-stage embryos may improve the chance of pregnancy by better matching the timing between the embryo and womb, and by helping to select embryos that are more likely to develop successfully. However, some people may lose embryos that would not survive to day 5 in the laboratory, even though they could have developed in the womb. We do not yet know if one strategy clearly leads to better outcomes over the full course of treatment, or if it affects pregnancy complications.

What did we want to find out?

We wanted to know the effects of transferring blastocyst-stage embryos versus cleavage-stage embryos on:

- the chance of having a live birth after using all embryos from one cycle (including fresh (new) embryos and embryos that have been frozen and then later thawed) or until a live birth occurs (called 'per cycle' or 'cumulative');
- the chance of having a live birth after one transfer (called 'per fresh transfer');
- the risk of complications during pregnancy or at birth (such as preterm birth);
- pregnancy rates;
- the risk of miscarriage;
- whether embryos are frozen for later use; and
- the chance that no embryo could be transferred.

What did we do?

We searched for studies known as 'randomised controlled trials' that compared blastocyst-stage and cleavage-stage embryo transfers in women undergoing IVF. We combined the studies' results and assessed our confidence in them. Our judgements are presented below in brackets.

What did we find?

- We included 36 studies involving 8389 women. Most studies took place in IVF settings and included participants with a good chance of pregnancy (i.e. they were generally younger and produced several embryos).

Chance of having a baby (live birth)

- Per IVF cycle, with fresh and frozen embryos (cumulative live birth rate): blastocyst transfer might slightly improve the chance of having a baby (low confidence).
- After a single fresh transfer: blastocyst transfer probably increases the chance of having a baby. For example, if 32 out of 100 women have a baby with cleavage-stage transfer, between 37 and 43 would have a baby with blastocyst-stage transfer (moderate confidence).

Preterm birth

- Blastocyst transfer probably increases the chance of having a baby born between 32 and 36 weeks (late preterm) (moderate confidence).

Chance of becoming pregnant ('clinical pregnancy')

- Blastocyst transfer may increase the chance of becoming pregnant after a fresh transfer (low confidence).

Miscarriage

- There may be little to no difference between blastocyst transfer and cleavage transfer in the miscarriage rate (low confidence).

Freezing of extra embryos

- Blastocyst transfer might reduce the chance of having embryos to freeze for later use (low confidence).

Failure-to-transfer rate

- Blastocyst transfer may increase the chance that no embryo successfully transfers (e.g. because no embryo develops to day 5) (low confidence).

What are the limitations of the evidence?

We are not fully confident in the results. Many studies were small, and there was variation in how the IVF treatment was given. People in the studies knew whether they were getting cleavage-stage or blastocyst-stage embryos, which may have affected the results. In some studies, most people had only one embryo transfer, so the results might be different if they were measured later once any frozen embryos had also been used. Most studies included women with a good chance of getting pregnant, so the findings may not apply to women with characteristics that make getting pregnant more difficult.

How current is this evidence?

The evidence is based on searches conducted up to October 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Blastocyst-stage versus cleavage-stage embryo transfer in assisted reproductive technology

Blastocyst-stage versus cleavage-stage embryo transfer in assisted reproductive technology

Population: women with subfertility
Settings: assisted reproductive technology clinics
Intervention: blastocyst-stage embryo transfer (day 5/6)
Comparison: cleavage-stage embryo transfer (day 2/3)

Outcomes per woman	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Cleavage-stage embryo transfer	**Blastocyst-stage embryo transfer**				
Cumulative live birth rate from fresh and frozen transfers (follow-up: 12 months)	618 per 1000	**668 per 1000** (627 to 704)	**OR 1.24** (1.04 to 1.47)	2328 (3 RCTs)	⊕⊕⊝⊝ Low[a,b]	A post hoc stratified analysis by the number of embryo transfers eliminated inconsistency across the studies, showing similar results when there was an average of less than two embryo transfers (OR 1.58, 95% CI 1.22 to 2.04), but little or no important difference when there was an average of two or more embryo transfers (OR 1.02, 95% CI 0.81 to 1.21) (Analysis 1.2). See Table 1 for differences in study variables.
Live birth rate per fresh transfer	322 per 1000	**398 per 1000** (369 to 426)	**OR 1.39** (1.23 to 1.56)	4787 (19 studies)	⊕⊕⊕⊝ Moderate[c]	When we conducted a sensitivity analysis removing studies at high overall risk of bias, we found a similar effect (OR 1.42, 95% CI 1.26 to 1.61). Subgroup analysis by the number of embryos transferred and prognosis was consistent with the primary analysis (Analysis 2.2 Analysis 2.3).
Cumulative preterm birth rate from fresh and frozen-thawed cycles (< 37 weeks)	35 per 1000	**58 per 1000** (39 to 85)	**OR 1.72** (1.14 to 2.59)	2194 (2 RCTs)	⊕⊕⊕⊝ Moderate[a]	Stratified analysis by early (< 32 weeks) and moderate/late preterm birth (32 to 37 weeks) showed similar results in moderate/late preterm birth (Analysis 3.3), but no important difference between transfer stages and more uncertainty due to wide confidence interval for early preterm birth (Analysis 3.2). Post hoc analysis of

(follow-up: 12 months)						cumulative preterm birth rate per live birth was consistent with the primary analysis (Analysis 3.4).
Clinical pregnancy rate per fresh transfer	389 per 1000	**455 per 1000** (433 to 478)	**OR 1.31** (1.20 to 1.44)	8389 (36 RCTs)	⊕⊕⊝⊝ Low[d]	Although moderate heterogeneity was observed when all studies were included in the meta-analysis (I^2 = 52%), a sensitivity analysis removing studies at high risk of bias yielded a similar effect estimate (OR 1.39, 95% CI 1.25 to 1.54; moderate-certainty evidence) with lower heterogeneity (I^2 = 45%). Therefore, we did not downgrade the certainty of evidence for inconsistency for this outcome.
Miscarriage rate per fresh transfer	77 per 1000	85 per 1000 (72 to 99)	**OR 1.12** (0.94 to 1.33)	6674 (25 RCTs)	⊕⊕⊝⊝ Low[d]	When we conducted sensitivity analysis by removing studies at high overall risk of bias, it resulted in a similar effect (OR 1.12, 95% CI 0.92 to 1.37; moderate-certainty evidence).
Embryo freezing rate	780 per 1000	609 per 1000 (574 to 648)	**OR 0.44** (0.38 to 0.52)	4620 (17 RCTs)	⊕⊕⊝⊝ Low[b,d]	When we conducted sensitivity analysis by removing studies at high overall risk of bias, it resulted in a similar effect (OR 0.42, 95% CI 0.35 to 0.51; moderate-certainty evidence).
Failure-to-transfer rate	20 per 1000	52 per 1000 (38 to 71)	**OR 2.69** (1.94 to 3.73)	5145 (21 RCTs)	⊕⊕⊝⊝ Low[d]	When we conducted sensitivity analysis by removing studies at high overall risk of bias, it resulted in a similar effect (OR 1.91, 95% CI 1.25 to 2.92; moderate-certainty evidence).

*The basis for the **assumed risk** is the median control group risk across studies. The **corresponding risk** (and its 95% CI) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial

GRADE Working Group grades of evidence
High certainty: further research is very unlikely to change our confidence in the estimate of effect.
Moderate certainty: further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.
Low certainty: further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
Very low certainty: we are very uncertain about the estimate.

[a]Downgraded one level for serious risk of bias: one study has some concerns of overall risk of bias due to deviations from the intended intervention
[b]Downgraded one level for inconsistency: substantial heterogeneity
[c]Downgraded one level for serious risk of bias: four studies have a high overall risk of bias (cumulative weight below 10%), while the rest have low overall risk of bias or some concerns. Sensitivity analysis is consistent with the primary analysis.
[d]Downgraded by two levels due to serious risk of bias: some studies have a high overall risk of bias (cumulative weight more than 20%), while the rest have low overall risk of bias or some concerns. Sensitivity analysis is consistent with the primary analysis.

BACKGROUND

Description of the condition

Worldwide, 15% of couples of reproductive age are affected by infertility [1]. The WHO (World Health Organization) estimates that, globally, between 48 million couples and 186 million individuals live with infertility [2]. Assisted reproductive technologies (ARTs), such as in vitro fertilisation (IVF), intracytoplasmic sperm injection (ICSI), and embryo freezing, are considered beneficial for many couples and women who are unlikely to conceive without infertility treatment, and for whom less invasive forms of treatment have failed or are unlikely to be effective.

The fledgling era of IVF, from 1980 to the mid-1990s, was characterised by relatively static successful pregnancy rates of around 20%. In the past decade, however, advances in ovarian stimulation, cell culture, embryo transfer, and new cryopreservation techniques, as well as freeze-all cycles, have culminated in significant overall improvements in the rate of successful pregnancies [3, 4]. This is evident in the annual statistical reports from different areas around the world. For example, one such report demonstrated a doubling of the pregnancy rate per embryo transfer cycle from 1994 to 2003, despite a decrease in the mean number of embryos transferred [5].

Prognosis in assisted reproductive technology (ART) varies considerably according to factors such as female age, ovarian reserve, and underlying infertility diagnosis. Consequently, treatment effects may differ across patient subgroups, making prognosis-based subgroup analysis a routine and essential component of ART research.

Description of the intervention and how it might work

In vitro fertilisation and embryo transfer

IVF involves the use of hormones to stimulate the ovaries to produce many eggs (oocytes), followed by egg collection (oocyte retrieval), addition or injection of sperm, fertilisation, embryo culture, and lastly, the return of a few selected embryos to the uterus (embryo transfer).

Cleavage-stage embryo transfer and early timing

Conventionally, two decades ago, embryos were transferred on either day 2 or day 3 after fertilisation when the embryos comprise two to eight cells, which is known as 'cleavage stage'. This was because the uterus was thought to provide the best environment for the survival of the embryo [6]. The question of optimal timing for embryo transfer arises from the differences between embryo development in IVF and the natural in vivo reproductive process. Day 2 is an early time point, at which morphological grading of the embryos is possible, allowing the embryologist to select the ‘best’ embryos for transfer. Embryo morphology, along with other factors, is thought to be highly indicative of pregnancy outcome [7]. Early placement in the uterus may be advantageous for the embryos because it limits the time spent in the in vitro environment of the embryology laboratory.

Shift towards blastocyst-stage embryo transfer

Over the past decade, there has been a shift in practice towards later embryo transfer, on day 5, 6, or 7, when embryos reach the blastocyst stage. Although blastocyst culture is not a novel concept, the first reported IVF pregnancy resulted from a transferred blastocyst. It was not adopted as standard practice early in the history of IVF because of low developmental rates beyond the cleavage stage and the observation that human embryos can survive when transferred prematurely into the uterus. [8, 9, 10, 11].

Embryo selection, implantation, and transfer policy

Extending embryo culture to the blastocyst stage may improve embryo selection by allowing embryos to demonstrate their full developmental potential in vitro. Improved selection could, in turn, lead to higher implantation rates following transfer. Higher implantation rates may influence clinical policies on the number of embryos transferred, as fewer embryos are typically needed to achieve pregnancy [12, 13]. Because the number of embryos transferred is a major determinant of both live birth and multiple pregnancy rates, differences in transfer policies across studies may contribute to heterogeneity in treatment effects. In particular, transferring more than one embryo can increase the chance of live birth but also substantially increases the risk of multiple pregnancy.

Rationale for evidence synthesis

The fact that many blastocyst-stage transfer trials are not prospectively randomised or are underpowered has contributed to the lack of a strong consensus about best practice for blastocyst culture. Therefore, an evidence-based approach using meta-analysis of small trials was needed to clarify the overall effects of blastocyst culture and to identify patient subgroups and clinical practices that might benefit most from this approach.

Advances in embryo culture systems

Interest in blastocyst culture increased substantially following advances in the understanding of embryo metabolic requirements and the introduction of more sophisticated culture systems, including sequential (stage-specific) media and co-culture techniques [14, 15, 16, 17]. A key finding was that cleavage-stage embryos and blastocysts have different metabolic needs and grow optimally in different in vitro environments. This led to the development of stage-specific media [18], such as the G1/G2 system described by Gardner in 1998 [19], in which embryos are transferred on day 3 from media with low glucose and limited amino acids to media with higher glucose concentrations and a broader amino acid profile[20]. With the use of stage-specific media [21], reported blastocyst development rates increased to as high as 60% to 65% [22]. More recently, the introduction of time-lapse systems, which allow continuous embryo assessment without removal from the incubator, has reduced the perceived necessity of sequential media while maintaining blastocyst development rates [23]. These technical advances prompted widespread investigation of blastocyst culture across clinics, resulting in numerous studies with mixed findings and ongoing debate regarding the benefits and drawbacks of extended embryo culture.

Biological rationale for blastocyst-stage transfer

There are two central arguments why blastocyst culture has possible advantages over traditional cleavage-stage transfer. Firstly, it is considered to be physiologically premature to expose early-stage embryos to the uterine environment, particularly one that has been subjected to superovulation and thus high levels of oestrogen [24]. In vivo, embryos travel through the fallopian tubes and do not reach the uterus before the morula (16-cell compacted) stage [25], which equates to at least day 4 of in vitro culture. The uterus provides a different nutritional environment from the oviduct. Therefore, it is postulated that the uterine environment may cause stress on the embryo if transferred at cleavage stage [26, 27], resulting in reduced implantation potential [20, 28].

Self-selection and implantation potential

The second argument for blastocyst-stage transfer is the reported higher implantation potential compared with cleavage-stage embryos. As a consequence of self-selection, it is postulated that only the most viable embryos are expected to develop into blastocysts. It is widely acknowledged that the morphological criteria used for selection of the best embryos on day 2 to 3 are limited. Many published studies that debate the correlation of morphological features with pregnancy rates can be found in the literature [29, 30, 31, 32, 33, 34]. It is now understood that a very large proportion of morphologically normal day-3 embryos are chromosomally abnormal or mosaic (with some normal and some abnormal cells), thus contributing to the 80% to 90% rate of implantation failure post-transfer that is observed in cleavage-stage protocols [35]. While the transfer of day 5 blastocysts cannot ensure the absence of chromosomal abnormality [36], Staessen and colleagues demonstrated that, at least in women older than 36 years, the incidence can be reduced from 59% in day 3 embryos to 35% in day 5 blastocysts [37]. The question that this review aims to answer is whether the higher implantation rates translate into higher live birth rates.

Potential disadvantages and patient subgroups

Arguments against blastocyst culture are largely related to this process of self-selection. Women undergoing blastocyst culture are expected to have a higher incidence of cycle cancellation due to failed embryo development [38, 39], and to have fewer embryos cryopreserved (frozen) [40, 41]. In patients with diminished ovarian reserve or fewer than three cleavage embryos, it is hard to assess the added-value effect of blastocyst stage transfer. In addition, women of advanced reproductive age often have diminished ovarian reserve and higher rates of aneuploidy and miscarriage, representing a distinct clinical subgroup. A recent observational study illustrated that blastocyst embryo transfer could lead to an equivalent pregnancy rate in a lower number of embryo transfers [42].

Measuring treatment effectiveness: cumulative live birth rate

A frozen–thawed cycle refers to an embryo transfer performed in a subsequent cycle, using embryos that were cryopreserved after oocyte retrieval and not transferred in the same cycle as the egg collection. Overall utilisation rates have previously been described as the total number of embryos transferred plus the number of embryos thawed, divided by the number of fertilised eggs. Whilst this approach presents information about the comparative number of pregnancy opportunities that each treatment approach can provide, it does not take into account the implantation potential for fresh and thawed embryos. The cumulative live birth rate is the only outcome that can assess this. An alternative efficacy formula was developed in the 2001 study by Schoolcraft and colleagues that does take cumulative pregnancy rate into account [22]. Using the formula (mean number of embryos transferred multiplied by implantation rate) plus (mean number of embryos cryopreserved multiplied by implantation rate) minus (1 minus cancellation rate), this group of researchers was able to demonstrate a 19% greater efficiency in blastocyst culture compared to cleavage-stage transfers. Disappointingly, such a utilisation and efficiency analysis is not possible in most RCTs due to the lack of reporting of frozen-thawed cycle outcomes within a reasonable time frame for trials. We would argue that a superior approach to both of the foregoing methods is to report the live birth rates for both fresh and frozen-thawed cycles following a single oocyte retrieval in women randomised to either cleavage-stage or blastocyst-stage transfers.

Why it is important to do this review

This is an update of a Cochrane review first published in 2002, and previously updated in 2005, 2007, 2012, 2016, and 2022 [43].

Advocates of blastocyst culture are confident that only the most viable embryos survive the extended culture to day 5 or 6. They argue that this results in a higher probability of implantation and requires fewer embryos to be transferred, thereby lowering the costly multiple birth rate [19, 44]. It is important to be aware that clinic policies may differ on the minimum criteria for blastocyst culture and the day on which this decision is made (for example, number of follicles, fertilised eggs, 8-cell embryos on day 3) [45]. It is also yet to be clarified if there are patient groups for whom blastocyst culture is disadvantageous. Most importantly, does blastocyst culture achieve the primary aim of providing the subfertile couple with a normal, healthy baby? Methods for identifying viable blastocysts are a popular research focus, involving a range of approaches which include identification of chromosomally-normal blastocysts by polar-body and blastomere, trophectoderm genetic analysis (using microarrays or next-generation sequencing, known as pre-implantation genetic screening [46]), and metabolomic analysis of culture media [47].

Critics of the blastocyst culture approach express concern at the increased incidence of women failing to have embryos available for transfer [38], although the day of participant recruitment into the blastocyst programme is crucial to this argument. Other negative outcomes reported to be associated with blastocyst culture include a higher incidence of monozygotic twinning and an altered sex ratio in favour of males [48, 49]. Monozygotic twinning is frequently reported at above 1% in ART cycles [50], whilst the background rate of monozygotic twins in spontaneous conceptions is in the order of 1 in 330. This twinning is associated with miscarriage, serious structural congenital anomalies, growth discrepancy, and twin-to-twin transfusion syndrome. An extended culture of an embryo has been implicated as one of the interventions associated with an increase in monozygotic twinning [51, 52, 53, 54], but a recent report suggests that improvements in cell culture techniques over time can result in a significant decrease in its incidence [55]. Similarly, as the underlying mechanisms that lead to an altered sex ratio are elucidated, whether it be media constituents or simply the morphological selection criteria [56], the imbalance may also be rectified.

This review aims to determine whether the number of days between oocyte retrieval and embryo transfer (that is, the embryo stage) has any effect on the success of ART treatment, and particularly, the live birth rate, the most important outcome for women undergoing treatment as well as for service providers.

See Glossary in Supplementary material 9.

OBJECTIVES

To assess the effects of blastocyst-stage (day 5 to 6) embryo transfer compared with cleavage-stage (day 2 to 3) embryo transfer on cumulative live birth rate per woman (defined as the occurrence of at least one live birth resulting from the fresh transfer and all subsequent frozen–thawed embryo transfers derived from the same oocyte retrieval), live birth rate per fresh transfer, and cumulative preterm birth outcomes using the same definition.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) [57] when conducting the review and PRISMA 2020 for the reporting [58]. We made the following changes to the protocol.

2026 update

- There has been a change in the authorship of this review since the previous published version. The byline has been updated for the 2026 review update in accordance with Cochrane authorship policies. In addition, we modified the title.
- We added cumulative live birth rate and cumulative preterm birth rate to the critical outcomes. As a consequence of this change, the objectives of the 2026 update were revised. This was approved by the Gynaecology and Fertility Group editors as an appropriate addition, given that the review already included cumulative pregnancy rate.
- We added ‘other cumulative obstetric and perinatal outcomes’ to the important outcomes. This was approved by the Gynaecology and Fertility Group editors as an appropriate addition, given that the review already included cumulative pregnancy rate.
- We eliminated multiple pregnancy rate as a primary adverse outcome because there are some other more critical outcomes reported in the new studies.
- We removed ‘cumulative pregnancy rate (cCPR) per woman (from both fresh and thawed cycles deriving from a single egg collection procedure)’ due to methodological concerns regarding interpretability across multiple transfers.
- Although not a change in methods per se, for clarification, we avoided subgroup analyses when only a few studies were available in each subgroup, as this provides insufficient statistical power, though for one critical outcome (cumulative live birth), we reported an exploratory, stratified analysis and did not present a test for subgroup differences.
- Disclosure of artificial intelligence use: artificial intelligence tools were used only for language editing (grammar and spelling). These tools did not generate content or influence any scientific judgments. All interpretations and conclusions are those of the authors.

2022 update

- We included a subgroup analysis comparing time-lapse system screening (TLS screening) on cleavage-stage transfer versus blastocyst-stage transfer (with and without time-lapse screening) because TLS was not available when the protocol was originally published, and the combination of stage of embryo and TLS could provide a different result.
- We assessed the risk of bias with the Cochrane risk of bias 2 (RoB 2) tool in line with updated Cochrane methods [59].
- We excluded cluster-RCTs because these types of trials are not appropriate in this context.

2016 update

- We added a post hoc subgroup analysis according to freezing technique to investigate statistical heterogeneity for the outcome cumulative pregnancy rate.
- We excluded women where frozen-thawed cycle results were shown, but no data were available from the fresh cycle, as studies without data from the initiating fresh cycle could not contribute to the cumulative analysis.

2012 update

- We shortened the objectives for clarity and focus.

2007 update

- We included only RCTs; we excluded quasi-RCTs as these are not valid in the context of fertility trials.
- We restricted subgroup and sensitivity analyses to specific clinical outcomes (live birth, clinical pregnancy, and multiple pregnancy) to prioritise clinically meaningful outcomes.

Criteria for considering studies for this review

Types of studies

We included only individually-randomised parallel-group trials (RCTs). We excluded quasi-RCTs (such as where allocation was by alternation, identification number, or day of the week) and cluster-randomised trials as these are not valid in the context of fertility trials. We also excluded cross-over trials unless pre-cross-over data were available.

Types of participants

Inclusion criteria

We included women affected by subfertility who were undergoing in vitro fertilisation (IVF) or intracytoplasmic sperm injection (ICSI) for therapeutic reasons or for oocyte donation. We included all patient prognosis groups, i.e. the categories to which women are assigned based on factors such as their age, type of subfertility, ovarian response to the superovulation drugs, and number of previous attempts. See the Investigation of heterogeneity and subgroup analysis section below for the categories.

Exclusion criteria

We excluded women whose IVF or ICSI cycle, or both, involved in vitro matured oocytes or pre-implantation genetic screening. We also excluded women whose frozen-thawed cycle results were shown, but where no data were available from the fresh cycle.

Types of interventions

Inclusion criteria

We included studies comparing blastocyst-stage (day 5 to 6) transfers to cleavage-stage (day 2 to 3) transfers in settings using single and sequential media culture methods for IVF and ICSI, where the embryos were grown for between two and six days in vitro prior to embryo transfer.

Exclusion criteria

We excluded studies using co-culture methods as an intervention.

We also excluded studies if they compared blastocyst-stage transfers to cleavage-stage transfers in frozen-thawed cycles but did not provide data from the fresh cycle.

Outcome measures

Critical outcomes

- Cumulative live birth rate (cLBR) per woman (number of live births after week 20 of pregnancy per woman) from both fresh and thawed cycles deriving from a single egg collection procedure during the follow-up period
- Live birth rate per woman (number of live births after week 20 of pregnancy per woman) following fresh transfer
- Cumulative preterm birth rate per woman (number of births between week 20 and week 37 of pregnancy per woman) from both fresh and thawed cycles deriving from a single egg collection procedure during the follow-up period

Important outcomes

- Clinical pregnancy rate per woman: number of women achieving a clinical pregnancy following fresh transfer (defined by the demonstration of foetal heart activity on ultrasound scan)
- Multiple-pregnancy rate per woman following fresh transfer: number of women with a multiple pregnancy (including twin and high-order multiple pregnancies, defined as three or more gestational sacs) following fresh embryo transfer
- Miscarriage rate for fresh transfer: number of occurrences per woman and per pregnant woman
- Embryo freezing rate per woman: number of women who had supernumerary embryos for transfer at a later date per woman.
- Failure-to-transfer rate (per woman): percentage of women who did not have an embryo transfer
- Other cumulative obstetric and perinatal outcomes (including both fresh and frozen-thawed cycles) during the follow-up period

Additional outcomes not appropriate for statistical pooling

We were unable to pool data per cycle or per embryo transfer, or per oocyte pick-up (OPU) [60]. If a study includes multiple cycles, transfers, or OPUs per woman, then results reported per cycle/transfer/OPU cannot be validly pooled. If the study reported results per cycle/transfer/OPU and this did not coincide with the point of randomisation (i.e. if the denominator did not coincide with the point in the treatment where randomisation occurred), we calculated new results using randomised participants as the denominator and treated excluded participants as having negative outcomes. However, due to the frequency with which this form of data is reported in the literature, we have entered them into the 'Data and analyses tables' for the following outcomes (Table 2).

- Implantation rate: the number of foetal sacs divided by the number of embryos transferred
- Blastocyst formation rate

Search methods for identification of studies

We obtained all reports that described (or might have described) RCTs comparing cleavage-stage embryo transfer and blastocyst-stage transfer in the treatment of subfertility using IVF or ICSI. We used no restrictions in terms of date, publication status, or language. We used the search strategies developed by the Information Specialist in the Cochrane Gynaecology and Fertility Group.

Electronic searches

On 1 October 2024, we searched the following databases. See Supplementary material 1 for our search strategies.

- The Cochrane Gynaecology and Fertility Group (CGF) Specialised Register of Controlled Trials, ProCite platform (searched on 1 October 2024 for trials from the inception of the Register to 23 November 2023 (the CGF Specialised Register has not been updated since November 2023))
- Cochrane Central Register of Controlled Trials (CENTRAL; 2024, Issue 9) via the Cochrane Register of Studies Online (CRSO), web platform (searched 1 October 2024)
- MEDLINE Ovid (searched from 1946 to 1 October 2024)
- Embase Ovid (searched from 1980 to 1 October 2024)
- PsycINFO Ovid (searched from 1806 to 1 October 2024)
- CINAHL EBSCO (Cumulative Index to Nursing and Allied Health Literature; searched from 1961 to 4 April 2020). Any later CINAHL search output is contained in the 2024 CENTRAL search output.

We combined the MEDLINE search with the Cochrane highly sensitive search strategy for identifying randomised trials, which appears in the *Cochrane Handbook for Systematic Reviews of Interventions* [61], hereafter referred to as the *Cochrane Handbook*.

We combined the Embase search with a trial filter developed by the Scottish Intercollegiate Guidelines Network (SIGN).

Searching other resources

We searched ClinicalTrials.gov (https://clinicaltrials.gov/) and the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP; https://apps.who.int/trialsearch) on 1 October 2024 (see Supplementary material 1).

We performed the search on titles, abstracts, and keywords of the listed articles.

We also searched the citation lists of relevant publications, review articles, and the included studies. We handsearched relevant conference abstracts and contacted experts in the field for any additional trials.

We checked Retraction Watch for retraction notices or expressions of concern relating to each study, and documented this in the characteristics table of each included study (see Supplementary material 2).

After the publication of this 2026 update, we intend to conduct a search for new trials biannually and update the review again as and when we find new trials to be incorporated.

Data collection and analysis

Review author Simone Cornelisse authored a study included in this review (Cornelisse 2024 [62, 63]). In accordance with Cochrane's policy, she did not participate in the study selection, data extraction, risk of bias assessment, or GRADE assessments for this study.

Selection of studies

Three review authors (DG, AC, SC) selected trials for inclusion in the review after employing the search strategy described previously. Two review authors independently viewed each record. We resolved any disagreements about eligibility through discussion. We have presented details of the included studies in the Supplementary material 2 tables, which provide a context for assessing the reliability of results. We have described the excluded studies in a table in the Supplementary material 3.

Data extraction and management

Two of three review authors (DG, AC, SC) extracted data from eligible studies using a data extraction form designed and pilot-tested by the authors. Two review authors independently viewed each record. We resolved any disagreements about data extraction through discussion.

We extracted the following information.

- Study characteristics: study design, setting, country, funding source, trial registration
- Participants: number randomised and analysed, inclusion/exclusion criteria, baseline characteristics (age, prognosis, infertility diagnosis, prior ART cycles, when available)
- Interventions: details of embryo culture and transfer protocols, number and stage of embryos transferred, use of fresh or frozen cycles, freezing technique
- Outcomes: definitions, measurement methods, and time points
- Risk of bias information: information relevant to the RoB 2 domains — (1) bias arising from the randomisation process, (2) bias due to deviations from intended interventions, (3) bias due to missing outcome data, (4) bias in measurement of the outcome, and (5) bias in selection of the reported result.

Where studies had multiple publications, we used the main trial report as the primary reference and derived additional details from secondary papers. We corresponded with study investigators for further data, as required.

Risk of bias assessment in included studies

Two review authors (DG and AC) independently assessed the included studies for risk of bias using the Cochrane risk of bias 2 (RoB 2) assessment tool [59, 64]. We assessed five domains of bias: bias arising from the randomisation process; bias due to deviations from intended interventions; bias due to missing outcome data; bias in measurement of the outcome; and bias in selection of the reported result. Our effect of interest was the effect of assignment to the intervention at baseline, regardless of whether the interventions were received as intended (the 'intention-to-treat effect'). If we had identified cross-over trials with usable pre-cross-over data, we would have assessed their risk of bias using the RoB 2 tool as for a parallel-group trial, considering only first-period data.

A potential deviation from the intended trial protocol is an adherence issue, such as moving to the cleavage transfer arm if there were few available embryos on day 3. Another potential deviation is a difference that could be found in the number of transferred embryos in each of the arms. Although both of these deviations could occur in the real world, they could also occur as a result of participating in a trial.

The outcomes we selected to be assessed for risk of bias are the same as those reported in our summary of findings table: cumulative live birth (fresh and frozen-thawed cycles), live birth, cumulative preterm birth (fresh and frozen-thawed cycles), clinical pregnancy, miscarriage, embryo freezing, and failure to transfer any embryo. We describe the measurement methods and time points in the Measures of treatment effect section.

We classified judgements as 'low' or 'high' risk of bias, or as causing 'some concerns', and assigned them as recommended in the *Cochrane Handbook*, Chapter 8 [61]. We resolved any disagreements about our risk of bias judgements through discussion. We described all judgements fully and presented the consensus judgements in the main review document (e.g. as a table, or a figure, or within a forest plot of the results).

These domain-level judgements provided the basis for our overall risk of bias judgement for the specific trial result being assessed.

We used the RoB 2 Excel tool (available on the riskofbiasinfo.org website) to process the use of RoB 2 and to store data for presentation [59]. We made the risk of bias data available by providing supplementary material.

We sought additional information from the principal author of trials that appeared to meet eligibility criteria but were unclear in aspects of methodology, or where the data were in a form that was unsuitable for meta-analysis. If a reply was not received within three weeks, we sent the trial author a reminder by re-sending the original email to the trial author.

Measures of treatment effect

For dichotomous data (e.g. clinical pregnancy rate), we expressed results for each study as odds ratios (ORs) with 95% confidence intervals (CIs) and combined them for meta-analysis in Cochrane's Review Manager software (RevMan) [65].

Unit of analysis issues

We conducted the primary analysis per woman randomised. We counted multiple live births (e.g. twins or triplets) as one live-birth event. Had we identified cross-over trials, we would have included only data from the first period, up to the point of cross-over, to avoid potential carry-over effects.

Dealing with missing data

We analysed the data on an intention-to-treat basis as far as possible (i.e. including all randomised participants in analysis, in the groups to which they were randomised). We attempted to obtain missing data from the original trialists. Where these were unobtainable, we imputed individual values for all outcomes by assuming that the outcome did not occur in participants with missing outcome data.

Reporting bias assessment

In view of the difficulty of detecting and correcting for publication bias and other reporting biases, we aimed to minimise their potential impact by ensuring that we conducted a comprehensive search for eligible studies and by being alert to duplication of data. We also searched for trial protocols and registrations (e.g. using ClinicalTrials.gov and WHO ICTRP) to compare prespecified outcomes and methods against those reported in the published manuscripts, in order to detect selective reporting. If there were 10 or more studies in an analysis, we generated a funnel plot to explore the possibility of small-study effects (i.e. a tendency for estimates of the intervention effect to be more beneficial in smaller studies).

Synthesis methods

We performed statistical analyses in accordance with the *Cochrane Handbook* [61]. The primary analyses included all studies. Before combining data from multiple studies, we considered whether the clinical and methodological characteristics of the included studies were sufficiently similar for meta-analysis to provide a clinically meaningful summary. We assessed statistical heterogeneity using the I^2 statistic. We interpreted an I^2 measurement greater than 50% to indicate substantial heterogeneity. We took statistical heterogeneity into account when interpreting the results.

We examined heterogeneity between the results of different studies by inspecting the scatter of data points, the overlap in their confidence intervals, and more formally, by checking the results of the Chi^2 tests. We pooled data for meta-analysis in RevMan [65], using the fixed-effect Mantel-Haenszel model. We entered the data on the graphs so that, for beneficial outcomes (e.g. pregnancy), data are displayed to the right of the line of no effect, and for detrimental outcomes (e.g. miscarriage), to the left of the line of no effect. A priori, we had planned to look at the possible contribution of differences in trial design to the heterogeneity identified.

Where possible, we pooled the outcomes statistically. We pooled data for meta-analysis in RevMan [65], using the fixed-effect Mantel-Haenszel model. We entered the data on the graphs so that, for beneficial outcomes (e.g. pregnancy), data are displayed to the right of the line of no effect, and for detrimental outcomes (e.g. miscarriage), to the left of the line of no effect. When meta-analysis was not possible due to substantial heterogeneity, insufficient data, or differences in outcome reporting, we presented the results in a narrative synthesis, structured by outcome and, where appropriate, by population or cointervention.

Investigation of heterogeneity and subgroup analysis

We conducted the subgroup analyses listed below for the following outcomes: cumulative live birth, live birth, clinical pregnancy, and multiple pregnancy. We conducted these analyses using the test for subgroup differences available in RevMan [65]. We avoided subgroup analyses when only a few studies were available in each subgroup, as this provides insufficient statistical power, though for one critical outcome (cumulative live birth), we reported the results of an exploratory, stratified analysis and did not present the test for subgroup differences.

- Studies that actively selected participants with a good prognosis (e.g. four or more zygotes, first two cycles, more than 10 follicles, young population, no male-factor individuals) versus participants with poor prognostic factors (e.g. previous failed ART cycles or poor response to ovulation stimulation) versus studies with unselected participants.
- Studies that randomised at the start of the cycle (i.e. prior to ovarian stimulation) were compared with those that randomised on the days immediately prior to and post-OPU (i.e. day of final ultrasound scan and prior to human chorionic gonadotropin trigger up to and including the day of fertilisation check, when numbers of oocytes are anticipated).
- Studies where the policy for the number of embryos placed was equal in both blastocyst-stage and cleavage-stage groups versus studies where fewer blastocyst-stage than cleavage-stage embryos were placed.

We planned to perform an overall assessment of risk of bias for subgroups.

Equity-related assessment

We did not examine any equity-related characteristics in this review because we believed it to be unlikely that studies would have reported data disaggregated by equity-relevant characteristics.

Sensitivity analysis

For our primary analysis, we pooled the data of all the studies. Then, for selected outcomes (cumulative live birth, live birth, cumulative preterm birth, clinical pregnancy, miscarriage, embryo freezing, and failure to transfer), we conducted sensitivity analyses by removing studies at overall high risk of bias.

Certainty of the evidence assessment

We prepared a summary of findings table using GRADEpro software and Cochrane and GRADE methods [61, 66]. We evaluated the overall certainty of the body of evidence for the main review outcomes (cumulative live birth rate, live birth rate, cumulative preterm birth, clinical pregnancy rate, miscarriage rate, embryo freezing rate, and failure to transfer any embryos rate) for the main review comparison (blastocyst-stage transfer versus cleavage-stage transfer). Review authors (DG, AC, SC) assessed the certainty of the evidence using the five GRADE criteria for downgrading: overall risk of bias (based on the RoB 2 tool assessment), inconsistency, imprecision, indirectness, and publication bias. Two review authors working independently made judgements about the certainty of the evidence (high, moderate, low, or very low), resolving any disagreements through discussion. We justified these judgements in the text of the review and in the footnotes of the summary of findings table.

Consumer involvement

Patients and other members of the public were not involved in this review due to limited resources, although the review authors did use core outcome sets for the review's outcomes; these core outcome sets were developed with consumer involvement.

RESULTS

Description of studies

See Supplementary material 2; Supplementary material 3; Supplementary material 5.

Results of the search

For the main searches conducted for the 2026 update of the review, we identified 2148 articles as potentially relevant for comparing blastocyst-stage versus cleavage-stage embryo transfer; we found no records from searching other sources. We excluded 2136 records as clearly irrelevant based on their title and/or abstract, and we retrieved 12 articles in full text. We included four new studies (five reports) (Clua 2022 [67, 68] (listed as awaiting classification in the 2022 update); Cornelisse 2024 (listed as ongoing in the 2022 update); Ma 2024 [69]; Mahmoudinia 2024 [70]), and excluded four new studies (six reports) in this update, which we added to the 'Characteristics of excluded studies tables' (Madani 2024 [71]; Mengels 2024 [72]; NCT04210414 [73] (listed as ongoing in the previous update); Torky 2021 [74, 75, 76]); two studies excluded in the previous update have been included as additional reports of included studies in this review (Bungum 2003 [77, 78]; Levron 2002 [79, 80]). We have one study awaiting classification (NCT01107002 [81]: listed as ongoing in the previous update; see Supplementary material 4). Including studies retrieved in the latest update, we now have 36 studies (49 articles) in the review. See Figure 1; Table 3; Table 4; Table 5.

Figure 1. Study flow diagram: results of search from review inception to 2025 (update of the flow diagram published in 2022)

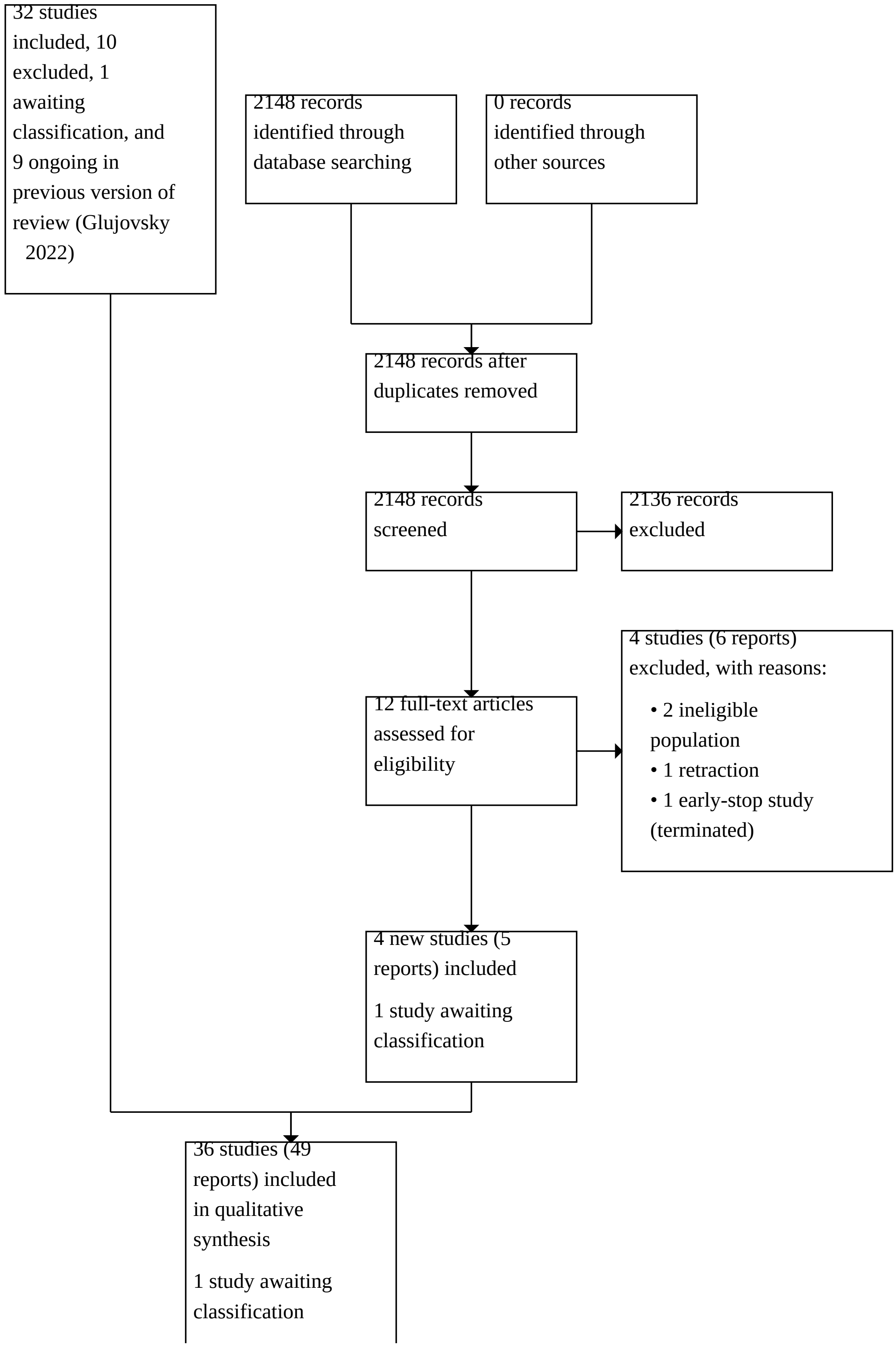


Figure 1. (Continued)

classification

5 ongoing studies

↓

36 studies included in quantitative synthesis (meta-analysis)

We also found five ongoing studies (from five records): one each from China (ChiCTR-ICR-15006184 [82]), the USA (Neuhausser 2020 [83]), Italy (ISRCTN48090543 [84]), Nigeria (PACTR202206898636627 [85]), and Egypt (PACTR201402000773124 [86]) (Supplementary material 5). One study previously listed as ongoing has been included as an additional report of Levi-Setti 2018 [87, 88].

Included studies

Study design and setting

We included 36 parallel-design RCTs in this review (8389 participants). The sample size of trials ranged from 20 women in Fisch 2007 [89] to 1202 women in Cornelisse 2024, including both comparison groups (Supplementary material 2).

Most trials (n = 31) were carried out in less than six months, other than the four largest trials (Cornelisse 2024; Kolibianakis 2004 [90]; Ma 2024 Yang 2018 [91, 92, 93]) and the one published by Clua 2022, which included a 12-month follow-up period.

All trials were reported to have been performed at single private clinics or university-based clinics, other than the multicentre study published by Cornelisse 2024, which was carried out in 21 hospitals and clinics in the Netherlands, and the multicentre study published by Ma 2024, which was carried out in 11 academic clinical centres in China.

Seventeen countries were represented in the included trials, with Belgium being the most prolific, providing six trials. The countries represented were: Australia (Livingstone 2002 [94]); Belgium (Devreker 2000 [95]; Emiliani 2003 [96]; Kolibianakis 2004; Papanikolaou 2005 [97, 98]; Papanikolaou 2006 [99, 100]; Van der Auwera 2002 [101]); Brazil (Motta 1998 [102]); China (Ma 2024; Yang 2018); Denmark (Bungum 2003); Egypt (Elgindy 2011 [103]; Gaafar 2015 [104]); France (Brugnon 2010 [105]); Greece (Pantos 2004 [106]); India (Kaur 2014 [107, 108]; Singh 2017 [109, 110]); Iran (Aziminekoo 2015 [111]; Mahmoudinia 2024); Israel (Coskun 2000 [112]; Levitas 2004 [113, 114]; Levron 2002); Italy (Levi-Setti 2018; Rienzi 2002 [115]; Schillaci 2002 [116]); Jordan (Karaki 2002 [117]); Netherlands (Cornelisse 2024); Spain (Clua 2022; Fernandez-Shaw 2015 [118, 119]; Ten 2011 [120]); Sweden (Hreinsson 2004 [121]); Turkey (Hatirnaz 2017 [122]); and the USA (Fisch 2007; Frattarelli 2003 [123]; Gardner 1998a [124]; Kaser 2017 [125, 126]).

Participants

Participant selection criteria comprised three main groups: unselected participants (Emiliani 2003; Fernandez-Shaw 2015; Gaafar 2015; Hatirnaz 2017; Karaki 2002; Kolibianakis 2004; Motta 1998; Pantos 2004; Schillaci 2002; Van der Auwera 2002); good prognostic factors, where participants were positively selected, that is, women were selected who would be expected to do well with blastocyst culture (Brugnon 2010; Bungum 2003; Clua 2022; Cornelisse 2024; Coskun 2000; Elgindy 2011; Fisch 2007; Frattarelli 2003; Gardner 1998a; Hreinsson 2004; Kaser 2017; Kaur 2014; Levi-Setti 2018; Levron 2002; Livingstone 2002; Ma 2024; Mahmoudinia 2024; Papanikolaou 2005; Papanikolaou 2006; Rienzi 2002; Singh 2017; Ten 2011; Yang 2018); and poor prognostic factors, where women were selected who had experienced multiple failures with conventional treatment or had a poor response to ovulation induction (Aziminekoo 2015; Devreker 2000; Levitas 2004). Most trials (n = 31) recruited women under 40 years of age, other than Cornelisse 2024, which included women under 44 years old, and Fernandez-Shaw 2015, Gardner 1998a, and Gaafar 2015, which had no age limit. Additionally, one study included women up to 50 years old, provided the oocytes came from younger donors (Clua 2022). The mean age across all the trials varied from 29 to 34 years.

We found only one study of participants using donor oocytes, which had a mean age of 26 years for donors and a mean age of 40 years for recipients (Clua 2022).

Interventions

Twenty-three trials used sequential media, of which 14 used Vitrolife G1/G2, while the remaining media were combinations of brands or made in-house. Six trials did not state the media used (Table 6).

Freezing of embryos in both experimental groups was reported in 17 of the 36 included trials (Brugnon 2010; Bungum 2003; Clua 2022; Cornelisse 2024; Fernandez-Shaw 2015; Gardner 1998a; Hreinsson 2004; Karaki 2002; Kolibianakis 2004; Levron 2002; Ma 2024; Motta 1998; Pantos 2004; Papanikolaou 2006; Rienzi 2002; Ten 2011; Van der Auwera 2002). Coskun 2000 reported no provision for day 5 freezing. Levitas 2004 stated that most of the remaining embryos were not suitable for freezing. Other interventions, such as assisted hatching, were either not provided or not reported on for most trials. Gardner 1998a was the only trial that practised assisted hatching, but only for the day 3 embryo transfer group.

For the cleavage-stage transfer groups, embryo transfers were made on day 3 in 29 trials; the transfers took place on day 2 in five trials (Devreker 2000; Emiliani 2003; Gaafar 2015; Motta 1998; Van der Auwera 2002); one study had a policy of transferring on day 2 or 3 (Levitas 2004); and one study had a policy of transferring frozen-thawed embryos on day 3 or 4 (Cornelisse 2024).

The trials that provided details on the ovarian stimulation regimen mostly reported using a similar gonadotropin-releasing hormone pituitary down-regulation protocol prior to human menopausal gonadotropin (HMG) and follicle stimulating hormone (FSH) administration. However, in some of the older trials and all the newest trials (Clua 2022; Cornelisse 2024; Kolibianakis 2004; Levi-Setti 2018; Ma 2024; Mahmoudinia 2024; Papanikolaou 2005; Papanikolaou 2006; Singh 2017), gonadotropin-releasing hormone antagonists were used.

Three trials evaluated the effects of adding time-lapse system (TLS) to cleavage-stage transfer (Clua 2022; Kaser 2017; Yang 2018).

Outcomes

- 19/36 trials reported live birth rate per fresh embryo transfer
- 3/36 trials reported cumulative live birth rate (fresh and frozen-thawed cycles)
- 2/36 trials reported cumulative preterm birth (fresh and frozen-thawed cycles)
- 36/36 trials reported clinical pregnancy rate
- 26/36 trials reported multiple pregnancy rate
- 14/36 trials reported high-order multiple pregnancy rate
- 25/36 trials reported miscarriage rate
- 17/36 trials reported embryo freezing rate (when there are supernumerary embryos for transfer at a later date)
- 21/36 trials reported failure-to-transfer rate
- 2/36 trials reported the following cumulative obstetrical and perinatal outcomes: major congenital anomalies, small for gestational age, and large for gestational age
- 1/36 trials reported the following cumulative obstetrical and perinatal outcomes: preeclampsia or eclampsia, preterm premature rupture of membranes, neonatal hospitalisation (> 3 days), and neonatal infection.

Contact with study authors

We attempted to contact study authors for information regarding methodology and outcome data. We received replies from 13 contact authors (Brugnon 2010, Bungum 2003; Clua 2022; Cornelisse 2024; Fernandez-Shaw 2015; Fisch 2007; Frattarelli 2003; Hreinsson 2004; Karaki 2002; Levitas 2004; Levron 2002; Papanikolaou 2005; Papanikolaou 2006; Rienzi 2002). Cumulative live birth data were provided by Fernandez-Shaw 2015. We did not receive any reply from Aziminekoo 2015 or Devreker 2000.

Excluded studies

In total, we excluded 12 studies (four of which were excluded in this update: Madani 2024; Mengels 2024; NCT04210414; Torky 2021). The reasons to exclude them were: five studies were not truly randomised (Cornelisse 2018 [127]; Green 2016 [128]; Holden 2017 [129]; Utsonomiya 2004 [130]; Zech 2007 [131, 132]); two studies used co-culture (Guerin 1991 [133]; Menezo 1992 [134]); in one study, fresh embryo transfers on day 5 or 6 were not the main intervention (Loup 2009 [135]); in two studies, only frozen embryo transfers were evaluated (Madani 2024; Mengels 2024); one study was retracted by the journal (Torky 2021); and one study was terminated (NCT04210414) (see Supplementary material 3).

Risk of bias in included studies

We attempted to obtain additional information regarding all aspects of randomisation, blinding, power analysis, and intention-to-treat from all trial authors (see Supplementary material 6).

We accessed the RoB 2 tool on 10 November 2024. We conducted risk of bias assessments for each outcome. These assessments, including all domain judgements and support for the judgements, are located within each included study's table of characteristics and at the side of all forest plots (Supplementary material 2; Supplementary material 6; Supplementary material 7). To access further detailed risk of bias assessment data, please use the following link.

Our risk of bias judgements for outcomes across all studies were predominantly 'some concerns' and 'high'. The most common reason for 'some concerns' of overall risk of bias was the lack of publication of the study's protocol in a trial registry, while the judgement of overall high risk of bias was mainly due to deviations from intended interventions, especially due to a higher number of transferred embryos.

Synthesis of results

Blastocyst-stage versus cleavage-stage embryo transfer

For an overview of our main analyses, please see Summary of findings 1 and Supplementary material 7 and Supplementary material 8.

Primary outcomes

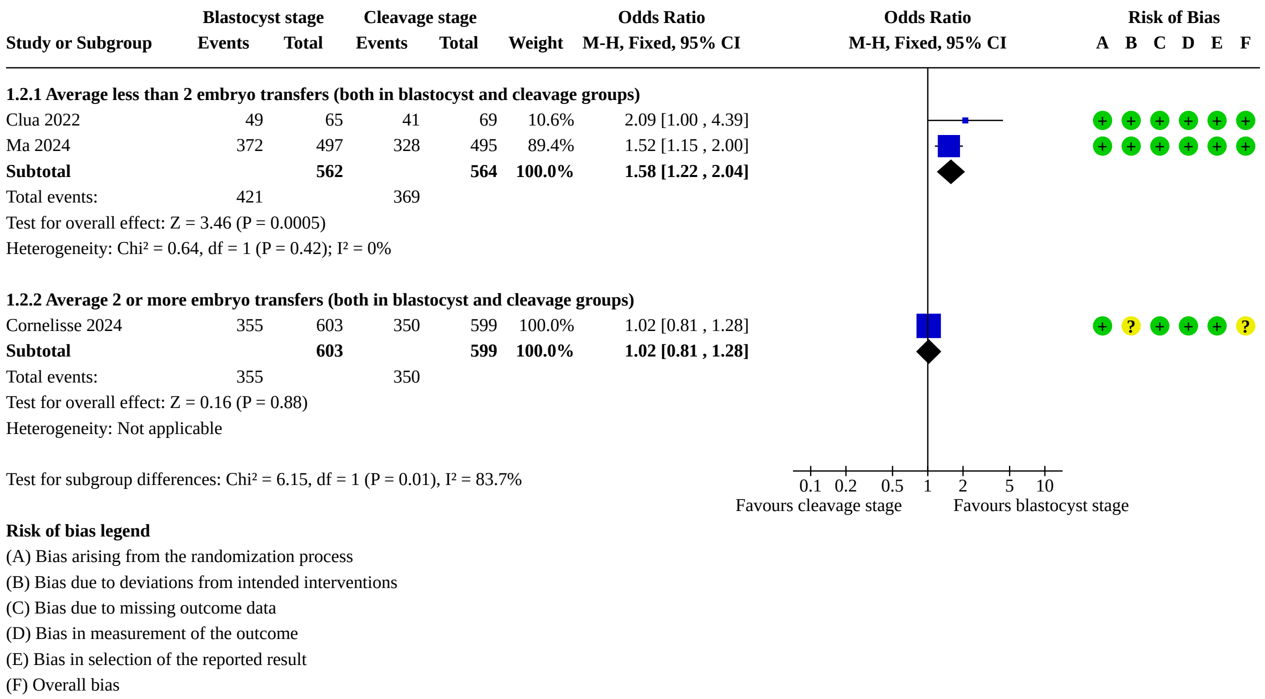
1. Cumulative live birth rate per woman (following fresh and frozen-thawed transfer)

Blastocyst-stage transfer may improve the cumulative clinical live birth rate in the first 12 months after oocyte retrieval (odds ratio (OR) 1.24, 95% confidence interval (CI) 1.04 to 1.47; I^2 = 70%; 3 studies, 2328 women; low-certainty evidence). However, this outcome showed substantial heterogeneity, with the largest study, which reported no significant differences between groups, accounting for 61% of the weight in the meta-analysis (Analysis 1.1 in Supplementary material 7).

This result suggests that if 62% of women achieved a live birth after fresh or frozen cleavage-stage transfer in the first 12 months after the oocyte retrieval, between 63% and 70% would do so after fresh or frozen blastocyst-stage transfer.

Notably, in the only study that reported similar results between both groups, the average number of transfers was two or higher in blastocyst and cleavage groups (1.99 versus 2.37), while in the other two studies, the average in both groups was significantly lower (1.51 versus 1.87; 1.46 versus 1.77). We conducted a post hoc stratified analysis according to the average number of embryo transfers (Analysis 1.2 in Supplementary material 7; Figure 2) and found evidence of a subgroup difference (test for subgroup differences: P = 0.01, I^2 = 83.7%). In the study with more embryo transfers per woman, there is probably little or no important difference between the groups (OR 1.02, 95% CI 0.81 to 1.28; 1 study, 1202 women; moderate-certainty evidence; Cornelisse 2024). Conversely, in those studies with a lower number of embryo transfers, the blastocyst group probably had a higher cLBR (OR 1.58, 95% CI 1.22 to 2.04; I^2 = 0%; 2 studies, 1126 women; moderate-certainty evidence; Clua 2022; Ma 2024). We also found differences in the women's ages, number of retrieved oocytes, and cryopreservation technique between Cornelisse 2024 and the other two studies (Clua 2022; Ma 2024).

Figure 2. Forest plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 1.2: cumulative live birth rate, grouped by average number of embryos (live birth per couple)



Subgroup and sensitivity analyses

We did not perform our planned subgroup analysis because the number of studies was insufficient to make them meaningful.

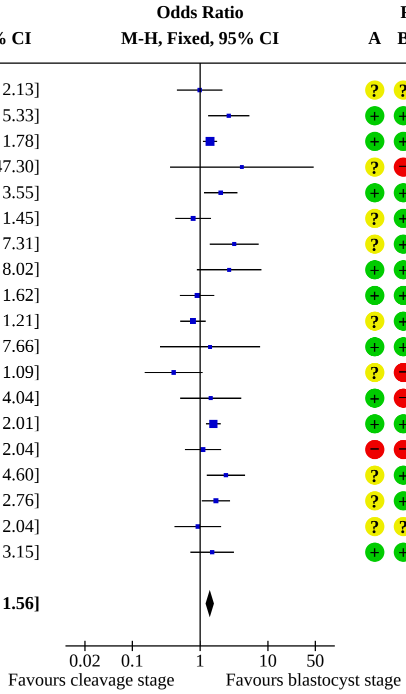
Our prespecified sensitivity analysis was not necessary as there were no studies at high overall risk of bias to be removed.

2. Live birth rate per woman after a fresh embryo transfer

The live birth rate per fresh embryo transfer was probably higher in the fresh blastocyst transfer group (OR 1.39, 95% CI 1.23 to 1.56; I^2 = 49%; 19 studies, 4787 women; moderate-certainty evidence; Analysis 2.1 in Supplementary material 7; Figure 3). This suggests that if 32% of women achieve a live birth after fresh cleavage-stage transfer, between 37% and 43% would do so after fresh blastocyst-stage transfer.

Figure 3. Forest plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 2.1: live birth rate following fresh transfer (live birth per couple)

Study or Subgroup	Blastocyst stage Events	Blastocyst stage Total	Cleavage stage Events	Cleavage stage Total	Weight	Odds Ratio M-H, Fixed, 95% CI	A	B	C	D	E	F
Brugnon 2010	22	55	21	52	2.8%	0.98 [0.45 , 2.13]	?	?	?	+	?	?
Clua 2022	37	65	23	69	2.1%	2.64 [1.31 , 5.33]	+	+	+	+	+	+
Cornelisse 2024	223	603	177	599	24.6%	1.40 [1.10 , 1.78]	+	+	+	+	+	+
Devreker 2000	3	11	1	12	0.2%	4.13 [0.36 , 47.30]	?	−	+	+	?	−
Elgindy 2011	52	100	35	100	3.7%	2.01 [1.14 , 3.55]	+	+	+	+	?	?
Emiliani 2003	33	82	41	89	5.2%	0.79 [0.43 , 1.45]	?	+	+	+	?	?
Fernandez-Shaw 2015	25	60	11	60	1.4%	3.18 [1.39 , 7.31]	?	+	+	+	?	?
Frattarelli 2003	15	29	8	28	0.9%	2.68 [0.89 , 8.02]	+	+	+	+	?	?
Hatirnaz 2017	33	101	35	100	5.2%	0.90 [0.50 , 1.62]	+	+	+	+	?	?
Levi-Setti 2018	54	194	64	194	10.2%	0.78 [0.51 , 1.21]	?	+	+	+	+	?
Levitas 2004	3	23	3	31	0.5%	1.40 [0.26 , 7.66]	+	+	+	+	?	?
Levron 2002	8	46	15	44	2.8%	0.41 [0.15 , 1.09]	?	−	+	+	?	−
Livingstone 2002	14	30	11	29	1.3%	1.43 [0.51 , 4.04]	+	−	+	+	?	−
Ma 2024	256	497	200	495	21.4%	1.57 [1.22 , 2.01]	+	+	+	+	+	+
Mahmoudinia 2024	27	120	25	120	4.3%	1.10 [0.60 , 2.04]	−	−	+	+	+	−
Papanikolaou 2005	38	80	23	84	2.6%	2.40 [1.25 , 4.60]	?	+	+	+	?	?
Papanikolaou 2006	56	175	38	176	5.7%	1.71 [1.06 , 2.76]	?	+	+	+	?	?
Rienzi 2002	24	50	24	48	2.8%	0.92 [0.42 , 2.04]	?	?	?	+	?	?
Van der Auwera 2002	24	70	17	66	2.5%	1.50 [0.72 , 3.15]	+	+	+	+	?	?
Total		**2391**		**2396**	**100.0%**	**1.39 [1.23 , 1.56]**						
Total events:	947		772									



Test for overall effect: Z = 5.33 (P < 0.00001)

Heterogeneity: Chi² = 35.63, df = 18 (P = 0.008); I² = 49%

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Subgroup and sensitivity analyses

We did not find clear evidence that the treatment effect differed between fresh cleavage-stage and blastocyst-stage transfer based either on the number of embryos transferred (test for subgroup differences: P = 0.49, I^2 = 0%; Analysis 2.2 in Supplementary material 7), the prognosis (test for subgroup differences: P = 0.55, I^2 = 0%; Analysis 2.3 in Supplementary material 7) or the randomisation day (test for subgroup differences: P = 0.12, I^2 = 48.2%; Analysis 2.4 in Supplementary material 7).

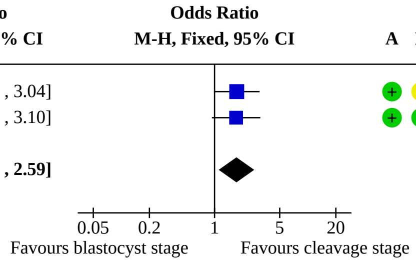
In a sensitivity analysis, we removed the studies at high overall risk of bias, and we found a higher live birth rate in the blastocyst group (OR 1.42, 95% CI 1.26 to 1.61; I^2 = 50%; 15 studies, 4375 women; moderate-certainty evidence; analysis not shown).

3. Cumulative preterm birth rate (following fresh and frozen-thawed cycles)

The cumulative preterm birth rate per woman after fresh and frozen cycles is probably higher in the blastocyst transfer group (OR 1.72, 95% CI 1.14 to 2.59; I^2 = 0%; 2 studies, 2194 women; moderate-certainty evidence; Analysis 3.1 in Supplementary material 7; Figure 4). This suggests that if 3.5% of women have a cumulative preterm birth after fresh or frozen cleavage-stage transfer in the first 12 months after the oocyte retrieval, between 3.9% and 8.5% would do so after fresh or frozen blastocyst-stage transfer.

Figure 4. Forest plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 3.1: cumulative preterm birth (< 37 weeks)

Study or Subgroup	Blastocyst stage Events	Blastocyst stage Total	Cleavage stage Events	Cleavage stage Total	Weight	Odds Ratio M-H, Fixed, 95% CI	Risk of Bias A	B	C	D	E	F
Cornelisse 2024	34	603	20	599	52.8%	1.73 [0.98 , 3.04]	+	?	+	+	+	?
Ma 2024	30	497	18	495	47.2%	1.70 [0.94 , 3.10]	+	+	+	+	+	+
Total		**1100**		**1094**	**100.0%**	**1.72 [1.14 , 2.59]**						
Total events:	64		38									



Test for overall effect: Z = 2.58 (P = 0.010)

Heterogeneity: Chi² = 0.00, df = 1 (P = 0.97); I² = 0%

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

We are uncertain about the evidence on early cumulative preterm birth as the confidence interval is very wide (OR 0.99, 95% CI 0.37 to 2.66; I^2 = 0%; 2 studies, 2194 women; Analysis 3.2 in Supplementary material 7).

The cumulative moderate to late preterm birth rate per woman after fresh and frozen cycles may be higher in the blastocyst transfer group (OR 1.97, 95% CI 1.07 to 3.65; 1 study, 1202 women; Analysis 3.3 in Supplementary material 7).

When we analysed the cumulative preterm birth rate per cumulative live birth, it may be higher in the blastocyst transfer group too (OR 1.63, 95% CI 1.08 to 2.47; I^2 = 0%; 2 studies, 1405 women; Analysis 3.4 in Supplementary material 7).

Subgroup and sensitivity analyses

Our prespecified sensitivity analysis was not necessary as there were no high-risk studies to be removed.

Secondary outcomes

4. Clinical pregnancy rate per woman

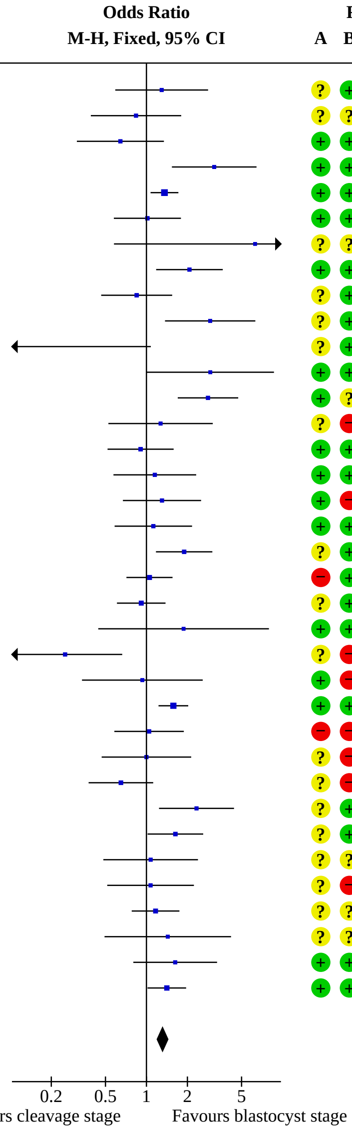
The clinical pregnancy rate for fresh embryos may be higher in the blastocyst transfer group (OR 1.31, 95% CI 1.20 to 1.44; I^2 = 52%; 36 studies, 8389 women; low-certainty evidence; Analysis 4.1 in Supplementary material 7; Figure 5). This suggests that if 39% of women achieved a clinical pregnancy after fresh cleavage-stage transfer, between 43% and 48% would do so after fresh blastocyst-stage transfer.

Figure 5. Forest plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 4.1: clinical pregnancy rate (per couple)

Study or Subgroup	Blastocyst stage Events	Blastocyst stage Total	Cleavage stage Events	Cleavage stage Total	Weight	Odds Ratio M-H, Fixed, 95% CI	A	B	C	D	E	F
Aziminekoo 2015	19	57	17	61	1.3%	1.29 [0.59 , 2.84]	?	+	+	+	?	?
Brugnon 2010	23	55	24	52	1.7%	0.84 [0.39 , 1.80]	?	?	?	+	+	?
Bungum 2003	32	61	36	57	2.1%	0.64 [0.31 , 1.34]	+	+	+	+	?	?
Clua 2022	46	65	30	69	1.0%	3.15 [1.54 , 6.44]	+	+	+	+	+	+
Cornelisse 2024	243	603	199	599	14.0%	1.36 [1.07 , 1.72]	+	+	+	+	+	+
Coskun 2000	39	100	39	101	2.8%	1.02 [0.58 , 1.79]	+	+	+	+	?	?
Devreker 2000	4	11	1	12	0.1%	6.29 [0.58 , 68.42]	?	?	+	+	−	−
Elgindy 2011	59	100	41	100	2.0%	2.07 [1.18 , 3.64]	+	+	+	+	?	?
Emiliani 2003	39	82	46	89	2.7%	0.85 [0.47 , 1.55]	?	+	+	+	?	?
Fernandez-Shaw 2015	31	60	16	60	0.9%	2.94 [1.37 , 6.31]	?	+	+	+	?	?
Fisch 2007	4	8	11	12	0.5%	0.09 [0.01 , 1.08]	?	+	+	+	?	?
Frattarelli 2003	18	29	10	28	0.5%	2.95 [1.00 , 8.65]	+	+	+	+	?	?
Gaafar 2015	76	126	44	126	2.1%	2.83 [1.70 , 4.72]	+	?	−	+	?	−
Gardner 1998a	32	45	31	47	1.0%	1.27 [0.53 , 3.07]	?	−	+	+	+	−
Hatirnaz 2017	42	101	44	100	3.0%	0.91 [0.52 , 1.58]	+	+	+	+	?	?
Hreinsson 2004	22	64	25	80	1.7%	1.15 [0.57 , 2.32]	+	+	+	+	?	?
Karaki 2002	28	80	24	82	1.8%	1.30 [0.67 , 2.52]	+	−	+	+	?	−
Kaser 2017	47	107	23	56	2.0%	1.12 [0.58 , 2.16]	+	+	+	+	+	+
Kaur 2014	66	150	44	150	2.9%	1.89 [1.17 , 3.05]	?	+	+	+	?	?
Kolibianakis 2004	75	226	75	234	5.8%	1.05 [0.71 , 1.56]	−	+	+	+	?	−
Levi-Setti 2018	71	194	75	194	5.6%	0.92 [0.61 , 1.38]	?	+	+	+	+	?
Levitas 2004	5	23	4	31	0.3%	1.88 [0.44 , 7.94]	+	+	+	+	?	?
Levron 2002	8	46	20	44	2.0%	0.25 [0.10 , 0.66]	?	−	+	+	?	−
Livingstone 2002	15	30	15	29	0.9%	0.93 [0.34 , 2.59]	+	−	+	+	?	−
Ma 2024	296	497	239	495	11.4%	1.58 [1.23 , 2.03]	+	+	+	+	+	+
Mahmoudinia 2024	30	120	29	120	2.6%	1.05 [0.58 , 1.88]	−	−	+	+	+	−
Motta 1998	21	58	21	58	1.6%	1.00 [0.47 , 2.13]	?	−	+	+	?	−
Pantos 2004	30	81	77	162	3.8%	0.65 [0.38 , 1.12]	?	−	+	+	?	−
Papanikolaou 2005	42	80	27	84	1.5%	2.33 [1.24 , 4.40]	?	+	+	+	?	?
Papanikolaou 2006	58	175	41	176	3.2%	1.63 [1.02 , 2.61]	?	+	+	+	?	?
Rienzi 2002	29	50	27	48	1.4%	1.07 [0.48 , 2.39]	?	?	?	+	?	?
Schillaci 2002	24	60	23	60	1.6%	1.07 [0.52 , 2.23]	?	−	+	+	?	−
Singh 2017	83	243	60	195	5.2%	1.17 [0.78 , 1.75]	?	?	−	+	?	−
Ten 2011	17	28	14	27	0.7%	1.44 [0.49 , 4.18]	?	?	−	+	?	−
Van der Auwera 2002	29	70	20	66	1.4%	1.63 [0.80 , 3.30]	+	+	+	+	?	?
Yang 2018	189	300	164	300	7.1%	1.41 [1.02 , 1.96]	+	+	+	+	+	+
Total		**4185**		**4204**	**100.0%**	**1.31 [1.20 , 1.44]**						
Total events:	1892		1636									

Test for overall effect: Z = 6.01 (P < 0.00001)

Heterogeneity: Chi² = 72.56, df = 35 (P = 0.0002); I² = 52%



Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Subgroup and sensitivity analyses

We did not find clear evidence that the treatment effect differed between fresh cleavage-stage and blastocyst-stage transfer based either on number of embryos transferred (test for subgroup differences: P = 0.16, I^2 = 44.7%; Analysis 4.2 in Supplementary material 7), or on the prognosis (test for subgroup differences: P = 0.55, I^2 = 0%; Analysis 4.3 in Supplementary material 7) or the randomisation day (test for subgroup differences: P = 0.20, I^2 = 36.1%; Analysis 4.4 in Supplementary material 7).

In a sensitivity analysis, we removed the studies at high overall risk of bias, and we found that the clinical pregnancy rate is probably higher in blastocyst transfer (OR 1.39, 95% CI 1.25 to 1.44; I^2 = 45%; 23 studies, 6039 women; analysis not shown).

5. Multiple pregnancy rate per woman

We are uncertain of the effect of the embryo stage on the multiple pregnancy rate in fresh cycles (OR 1.11, 95% CI 0.91 to 1.35; I^2 = 22%; 26 studies, 6776 women; Analysis 5.1 in Supplementary material 7).

This suggests that if 6% of women have a multiple pregnancy after fresh cleavage-stage transfer, between 6% and 8% would do so after fresh blastocyst-stage transfer.

Subgroup and sensitivity analyses

In the subgroup analysis by the number of transferred embryos, the test for subgroup differences was P = 0.006, I^2 = 80.2%. The multiple pregnancy may be higher in blastocyst group if the same number of embryos are transferred (OR 1.25, 95% CI 0.99 to 1.58; I^2 = 9%; 16 studies, 4009 women), and lower when more embryos are transferred in blastocyst stage (OR 0.57, 95% CI 0.37 to 0.90; I^2 = 0%; 7 studies, 824 women). See Analysis 5.2 in Supplementary material 7.

We did not find evidence that the treatment effect differed between fresh cleavage-stage and blastocyst-stage transfer based on the prognosis (test for subgroup differences: P = 0.97, I^2 = 0%; Analysis 5.3 in Supplementary material 7).

In the subgroup analysis by the randomisation day, the test for subgroup differences was P = 0.003, I^2 = 78.8%. We are uncertain if the multiple pregnancy differs according to the blastocyst or cleavage group in all except those cases in which randomisation was performed on day 2/3, when multiple pregnancy is probably higher in blastocyst group (OR 1.76, 95% CI 1.20 to 2.58; I^2 = 45%; 5 studies, 1674 women) (Analysis 5.4 in Supplementary material 7).

We are uncertain of the effect of the embryo stage transfer on the high-order multiple pregnancy rate in fresh cycles (OR 0.44, 95% CI 0.18 to 1.08; I^2 = 0%; 14 studies, 2575 women) (Analysis 5.5 in Supplementary material 7).

In a sensitivity analysis, we removed the studies at high overall risk of bias, and we found that transferring at blastocyst stage probably increases the multiple pregnancy rate (OR 1.39, 95% CI 1.10 to 1.76; 18 studies, 5538 women; analysis not shown).

6. Miscarriage rate for fresh transfer

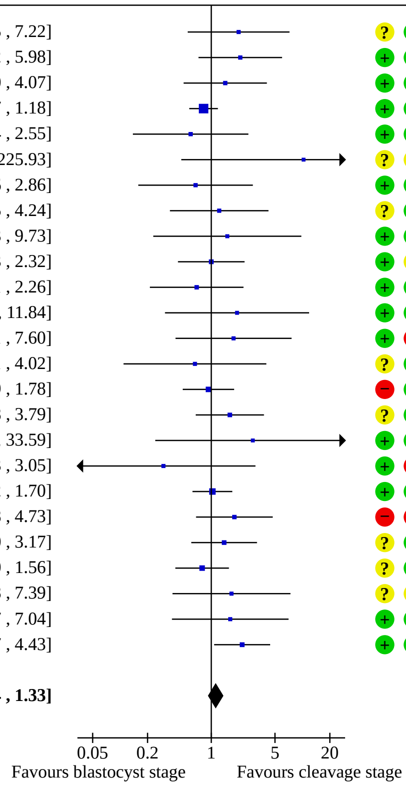
There may be minimal to no difference between groups in miscarriage rate (OR 1.12, 95% CI 0.94 to 1.33; I^2 = 0%; 25 studies, 6674 women; low-certainty evidence; Analysis 6.1 in Supplementary material 7; Figure 6). This suggests that if 8% of women have a miscarriage after fresh cleavage-stage transfer, between 7% and 10% would do so after fresh blastocyst-stage transfer.

Figure 6. Forest plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 6.1: miscarriage

Study or Subgroup	Blastocyst stage Events	Blastocyst stage Total	Cleavage stage Events	Cleavage stage Total	Weight	Odds Ratio M-H, Fixed, 95% CI	A	B	C	D	E	F
Aziminekoo 2015	7	57	4	61	1.5%	2.00 [0.55 , 7.22]	?	+	+	+	?	?
Bungum 2003	12	61	6	57	2.1%	2.08 [0.72 , 5.98]	+	+	+	+	?	?
Clua 2022	9	65	7	69	2.5%	1.42 [0.50 , 4.07]	+	+	+	+	+	+
Cornelisse 2024	61	603	72	599	28.0%	0.82 [0.57 , 1.18]	+	+	+	+	+	+
Coskun 2000	3	100	5	101	2.1%	0.59 [0.14 , 2.55]	+	+	+	+	?	?
Devreker 2000	3	11	0	12	0.1%	10.29 [0.47 , 225.93]	?	?	+	+	+	?
Elgindy 2011	4	59	4	41	1.9%	0.67 [0.16 , 2.86]	+	+	+	+	?	?
Fernandez-Shaw 2015	6	60	5	60	1.9%	1.22 [0.35 , 4.24]	?	+	+	+	?	?
Frattarelli 2003	3	29	2	28	0.8%	1.50 [0.23 , 9.73]	+	+	+	+	?	?
Gaafar 2015	12	126	12	126	4.7%	1.00 [0.43 , 2.32]	+	?	−	+	?	−
Hatirnaz 2017	5	101	7	100	2.9%	0.69 [0.21 , 2.26]	+	+	+	+	?	?
Hreinsson 2004	3	64	2	80	0.7%	1.92 [0.31 , 11.84]	+	+	+	+	?	?
Karaki 2002	5	80	3	82	1.2%	1.76 [0.41 , 7.60]	+	−	+	+	?	−
Kaur 2014	2	150	3	150	1.3%	0.66 [0.11 , 4.02]	?	+	+	+	?	?
Kolibianakis 2004	19	226	21	234	8.1%	0.93 [0.49 , 1.78]	−	+	+	+	?	−
Levi-Setti 2018	14	194	9	194	3.6%	1.60 [0.68 , 3.79]	?	+	+	+	+	?
Levitas 2004	2	23	1	31	0.3%	2.86 [0.24 , 33.59]	+	+	+	+	?	?
Livingstone 2002	1	30	3	29	1.3%	0.30 [0.03 , 3.05]	+	−	+	+	?	−
Ma 2024	33	497	32	495	12.9%	1.03 [0.62 , 1.70]	+	+	+	+	+	+
Mahmoudinia 2024	12	120	7	120	2.7%	1.79 [0.68 , 4.73]	−	−	+	+	+	−
Papanikolaou 2005	15	80	12	84	4.1%	1.38 [0.60 , 3.17]	?	+	+	+	?	?
Papanikolaou 2006	17	175	21	176	8.1%	0.79 [0.40 , 1.56]	?	+	+	+	?	?
Rienzi 2002	5	50	3	48	1.2%	1.67 [0.38 , 7.39]	?	?	?	+	?	?
Van der Auwera 2002	5	70	3	66	1.2%	1.62 [0.37 , 7.04]	+	+	+	+	?	?
Yang 2018	25	300	12	300	4.7%	2.18 [1.07 , 4.43]	+	+	+	+	+	+
Total		**3331**		**3343**	**100.0%**	**1.12 [0.94 , 1.33]**						
Total events:	283		256									

Test for overall effect: Z = 1.23 (P = 0.22)

Heterogeneity: Chi² = 19.04, df = 24 (P = 0.75); I² = 0%



Risk of bias legend

(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Seven of the included trials reported on the presence or absence of monozygotic twinning. A total of three sets of monozygotic twins were reported: two with cleavage-stage embryo transfers and only one set of monozygotic twins from blastocyst transfer.

When we analysed the miscarriage rate per clinical pregnancy in a fresh embryo transfer (Analysis 6.2 in Supplementary material 7), we were uncertain about any important differences (OR 0.87, 95% CI 0.72 to 1.06; I^2 = 19%; 25 studies, 2850 women).

Subgroup and sensitivity analyses

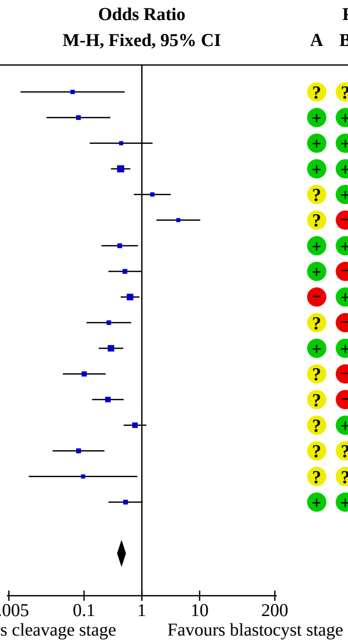
In a sensitivity analysis, we removed the studies at high overall risk of bias and found little or no evidence of a difference in miscarriage rate (OR 1.12, 95% CI 0.92 to 1.37; 20 studies, 5501 women; analysis not shown).

7. Embryo freezing rate per woman

Rates of embryo freezing when there are supernumerary embryos for transfer at a later date per woman may be lower in the blastocyst transfer group (OR 0.44, 95% CI 0.38 to 0.52; I^2 = 82%; 17 studies, 4620 women; low-certainty evidence; Analysis 7.1 in Supplementary material 7; Figure 7). Although we are uncertain about the result, the direction of effect is consistent in most studies. This suggests that if 78% of women have embryo freezing after fresh cleavage-stage transfer, between 57% and 65% would do so after fresh blastocyst-stage transfer.

Figure 7. Forest plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 7.1: embryo freezing rate

Study or Subgroup	Blastocyst stage Events	Blastocyst stage Total	Cleavage stage Events	Cleavage stage Total	Weight	Odds Ratio M-H, Fixed, 95% CI	A	B	C	D	E	F
Brugnon 2010	42	55	51	52	2.4%	0.06 [0.01 , 0.50]	?	?	?	+	?	?
Bungum 2003	36	61	54	57	4.4%	0.08 [0.02 , 0.28]	+	+	+	+	?	?
Clua 2022	57	65	65	69	1.5%	0.44 [0.13 , 1.53]	+	+	+	+	+	+
Cornelisse 2024	513	603	557	599	15.9%	0.43 [0.29 , 0.63]	+	+	+	+	+	+
Fernandez-Shaw 2015	39	60	33	60	2.2%	1.52 [0.73 , 3.17]	?	+	+	+	?	?
Gardner 1998a	29	45	14	47	0.9%	4.27 [1.78 , 10.24]	?	−	+	+	+	−
Hreinsson 2004	15	64	34	80	4.4%	0.41 [0.20 , 0.86]	+	+	+	+	?	?
Karaki 2002	22	80	35	82	4.8%	0.51 [0.26 , 0.98]	+	−	+	+	?	−
Kolibianakis 2004	114	226	145	234	13.5%	0.62 [0.43 , 0.91]	−	+	+	+	?	−
Levron 2002	12	46	25	44	3.6%	0.27 [0.11 , 0.65]	?	−	+	+	?	−
Ma 2024	426	497	472	495	12.9%	0.29 [0.18 , 0.48]	+	+	+	+	+	+
Motta 1998	15	58	45	58	6.4%	0.10 [0.04 , 0.24]	?	−	+	+	?	−
Pantos 2004	16	81	79	162	8.1%	0.26 [0.14 , 0.48]	?	−	+	+	?	−
Papanikolaou 2006	115	175	126	176	8.2%	0.76 [0.48 , 1.20]	?	+	+	+	?	?
Rienzi 2002	18	50	42	48	5.2%	0.08 [0.03 , 0.23]	?	?	?	+	?	?
Ten 2011	20	28	26	27	1.4%	0.10 [0.01 , 0.83]	?	?	−	+	?	−
Van der Auwera 2002	26	70	35	66	4.3%	0.52 [0.26 , 1.04]	+	+	+	+	?	?
Total		**2264**		**2356**	**100.0%**	**0.44 [0.38 , 0.52]**						
Total events:	1515		1838									



Test for overall effect: Z = 10.54 (P < 0.00001)

Heterogeneity: Chi² = 87.06, df = 16 (P < 0.00001); I² = 82%

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

In a sensitivity analysis, we removed the studies at high overall risk of bias, and we found that the embryo freezing rate is lower in blastocyst transfers (OR 0.42, 95% CI 0.35 to 0.51; 10 studies, 3402 women; analysis not shown).

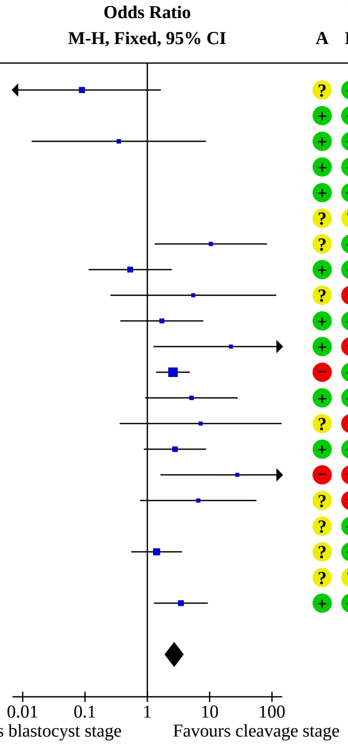
8. Failure-to-transfer rate

Rates of failure to transfer any embryos may be higher in the blastocyst transfer group (OR 2.69, 95% CI 1.94 to 3.73; I^2 = 35%; 21 studies, 5145 women; low-certainty evidence; Analysis 8.1 in Supplementary material 7; Figure 8). This suggests that if 2% of women have a failure to transfer embryos after fresh cleavage-stage transfer, between 4% and 7% would do so after fresh blastocyst-stage transfer.

Figure 8. Forest plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 8.1: failure to transfer any embryos (per couple)

Study or Subgroup	Blastocyst stage Events	Blastocyst stage Total	Cleavage stage Events	Cleavage stage Total	Weight	Odds Ratio M-H, Fixed, 95% CI	Risk of Bias A	B	C	D	E	F
Aziminekoo 2015	0	57	5	61	11.0%	0.09 [0.00 , 1.65]	?	+	+	+	?	?
Bungum 2003	0	61	0	57		Not estimable	+	+	+	+	?	?
Clua 2022	0	65	1	69	3.0%	0.35 [0.01 , 8.71]	+	+	+	+	+	+
Cornelisse 2024	0	603	0	599		Not estimable	+	+	+	+	+	+
Coskun 2000	0	100	0	101		Not estimable	+	+	+	+	?	?
Devreker 2000	0	11	0	12		Not estimable	?	?	+	+	?	?
Emiliani 2003	10	99	1	94	1.9%	10.45 [1.31 , 83.31]	?	+	+	+	?	?
Frattarelli 2003	3	29	5	28	9.5%	0.53 [0.11 , 2.47]	+	+	+	+	?	?
Gardner 1998a	2	45	0	47	1.0%	5.46 [0.25 , 116.92]	?	−	+	+	+	−
Hreinsson 2004	4	64	3	80	5.2%	1.71 [0.37 , 7.94]	+	+	+	+	?	?
Karaki 2002	9	80	0	82	0.9%	21.92 [1.25 , 383.34]	+	−	+	+	?	−
Kolibianakis 2004	36	226	16	234	27.6%	2.58 [1.39 , 4.80]	−	+	+	+	?	−
Levitas 2004	6	23	2	31	2.6%	5.12 [0.93 , 28.26]	+	+	+	+	?	?
Levron 2002	3	46	0	44	1.0%	7.16 [0.36 , 142.76]	?	−	+	+	?	−
Ma 2024	11	497	4	495	8.2%	2.78 [0.88 , 8.79]	+	+	+	+	+	+
Mahmoudinia 2024	12	120	0	120	0.9%	27.76 [1.62 , 474.54]	−	−	+	+	+	−
Motta 1998	6	58	1	58	1.9%	6.58 [0.77 , 56.47]	?	−	+	+	?	−
Papanikolaou 2005	0	80	0	84		Not estimable	?	+	+	+	?	?
Papanikolaou 2006	11	175	8	176	15.6%	1.41 [0.55 , 3.59]	?	+	+	+	?	?
Rienzi 2002	0	50	0	48		Not estimable	?	?	?	+	?	?
Van der Auwera 2002	18	70	6	66	9.6%	3.46 [1.28 , 9.37]	+	+	+	+	?	?
Total		**2559**		**2586**	**100.0%**	**2.69 [1.94 , 3.73]**						
Total events:	131		52									

Test for overall effect: Z = 5.96 (P < 0.00001)



Heterogeneity: Chi² = 21.63, df = 14 (P = 0.09); I² = 35%

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

In a sensitivity analysis, we removed the studies at high overall risk of bias. We found that the failure-to-transfer rate was higher in blastocyst transfers (OR 1.91, 95% CI 1.25 to 2.92; 15 studies, 3985 women; analysis not shown).

9. Other cumulative obstetric and perinatal outcomes (following fresh and frozen-thawed cycles)

In one study (Ma 2024), the cumulative preeclampsia or eclampsia rates per woman after fresh and frozen cycles was higher in the cleavage transfer group (OR 2.86, 95% CI 1.02 to 8.01; 1 study, 992 women; Analysis 9.1, in Supplementary material 7).

On the other hand, in the same study, the cumulative preterm premature rupture of membrane rate (OR 3.22, 95% CI 1.44 to 7.22; 1 study, 992 women; Analysis 9.2, in Supplementary material 7), the cumulative neonatal hospitalisation > 3 days rate (OR 1.94, 95% CI 1.23 to 3.06; 1 study, 992 women; Analysis 9.3 in Supplementary material 7), and the cumulative neonatal infection rate (OR 2.23, 95% CI 1.08 to 4.61; 1 study, 992 women; Analysis 9.4 in Supplementary material 7) per woman after fresh and frozen cycles were higher in the blastocyst transfer group (Ma 2024).

The evidence is insufficient to evaluate cumulative major congenital anomalies per woman after fresh and frozen cycles; the confidence interval is very wide (OR 1.15, 95% CI 0.54 to 2.43; I^2 = 0%; 2 studies, 2194 women; Analysis 9.5 in Supplementary material 7).

No clear evidence of a difference was found for cumulative small for gestational age (OR 0.99, 95% CI 0.67 to 1.48; I^2 = 0%; 2 studies, 2194 women; Analysis 9.6 in Supplementary material 7) and large for gestational age (OR 1.09, 95% CI 0.80 to 1.48; I^2 = 0%; 2 studies, 2194 women; Analysis 9.7 in Supplementary material 7) per woman after fresh and frozen cycles.

For the rest of the cumulative obstetric and perinatal outcomes that were assessed, see Table 7.

Our prespecified sensitivity analysis was not necessary as there were no high-risk studies to be removed.

10. Other data

Blastocyst formation rates

As reported in Table 2, blastocyst formation rates, which show the proportion of two pronuclei (2PN) embryos that get to blastocyst stage (day 5 to 6 transfer only) ranged from 22.4% in Aziminekoo 2015 to 97% in Elgindy 2011.

Implantation data

For blastocyst-stage transfer, the implantation rate varied from 14.5% to 72.5%, and for cleavage-stage transfer, the implantation rate varied from 2.9% to 92% (see Table 2).

Reporting biases

We generated funnel plots for the outcomes of live birth and clinical pregnancy. They did not suggest publication bias (see Figure 9 and Figure 10).

Figure 9. Funnel plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 2.1: live birth rate following fresh transfer (live birth per couple)

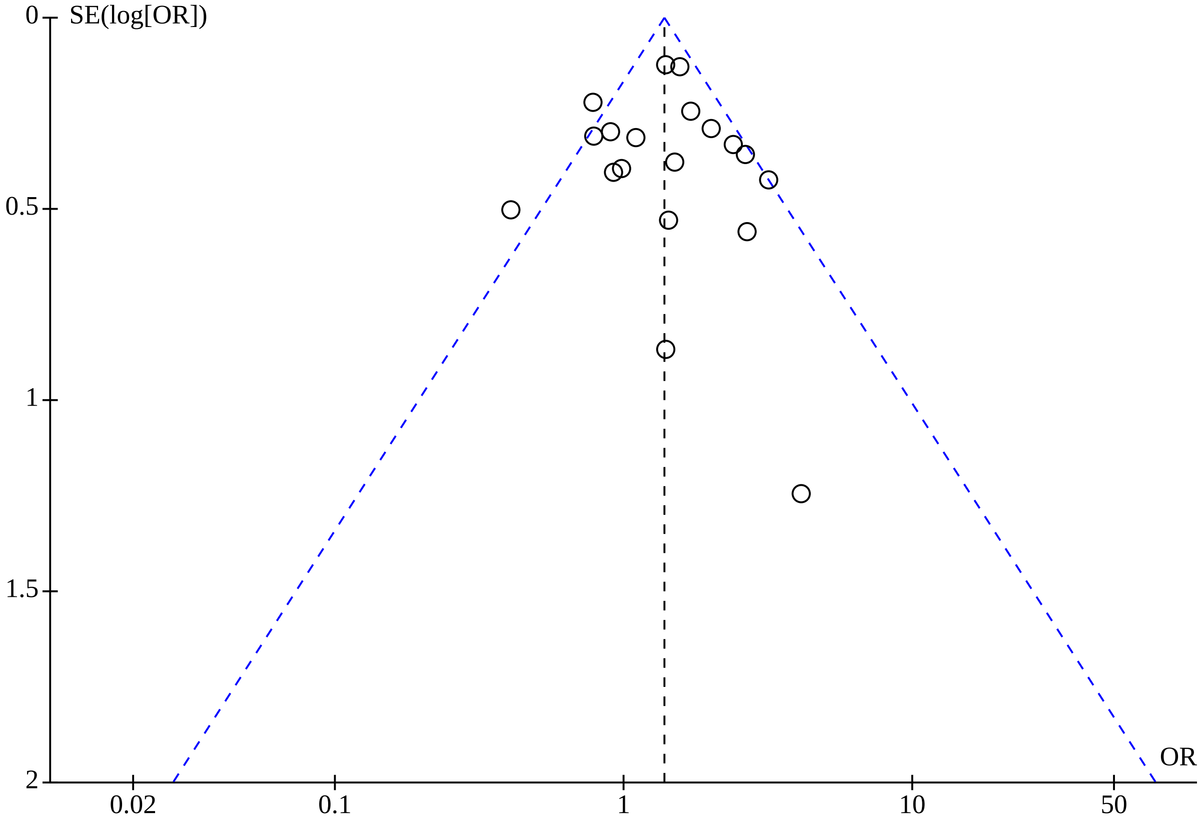
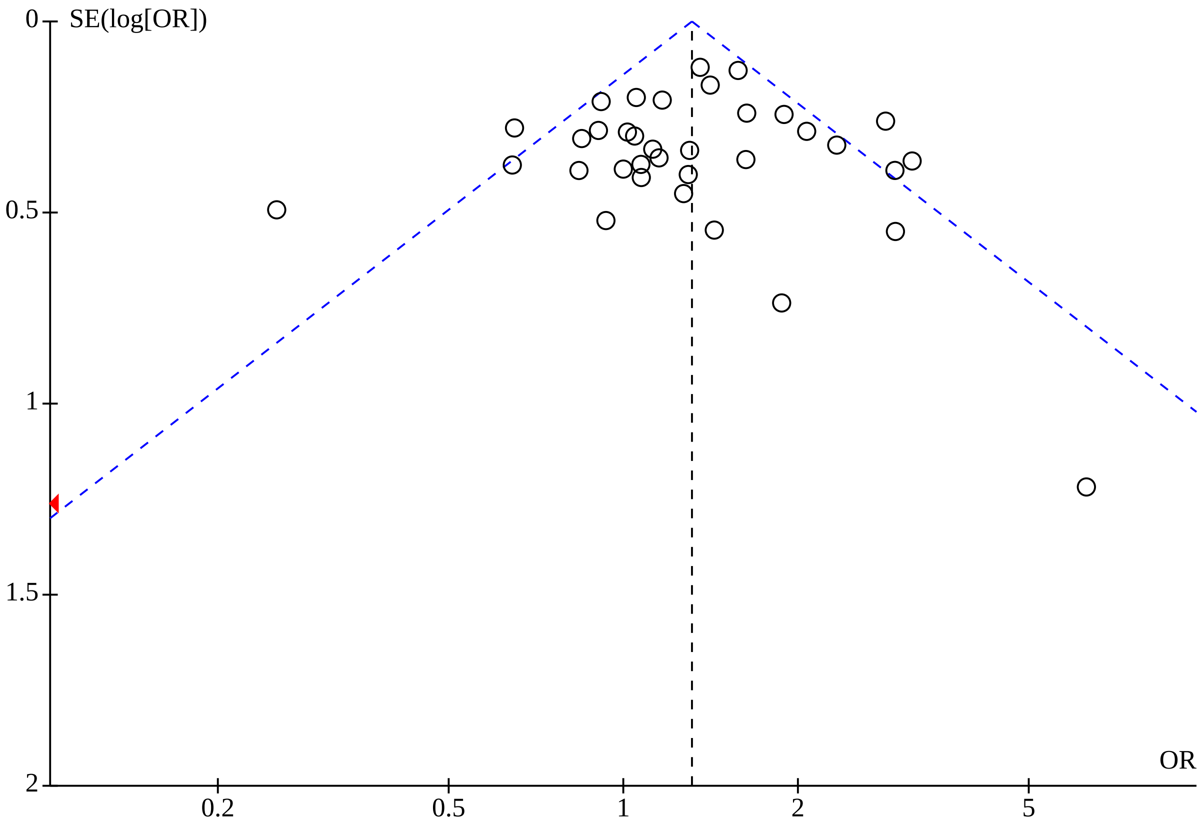


Figure 10. Funnel plot of comparison, blastocyst- versus cleavage-stage transfer, outcome 4.1: clinical pregnancy rate (per couple)



DISCUSSION

Summary of main results

This updated review includes 36 randomised controlled trials that included 8389 women (most of whom had a good prognosis for pregnancy) and compared embryo transfer at blastocyst stage versus at cleavage stage.

From the three trials that reported cumulative live birth rate (including both fresh and frozen-thawed cycles) (Clua 2022; Cornelisse 2024; Ma 2024), there is low-certainty evidence that suggests a possible benefit for blastocyst-stage embryo transfer over cleavage-stage transfer, but with significant heterogeneity. In the 19 trials that reported live birth rates after fresh transfer, which mainly recruited women with a good prognosis, there was moderate-certainty evidence of a probable benefit for blastocyst transfer. Clinical pregnancy rates from fresh embryos, also based on studies of women with a good prognosis, may be higher with blastocyst-stage transfer (low-certainty evidence). However, cumulative preterm birth from fresh and frozen-thawed cycles is probably more likely with blastocyst transfer, particularly for moderate/late preterm birth (moderate-certainty evidence).

Regarding the heterogeneity in our analysis, although all three studies reporting cumulative live birth rate followed participants for 12 months, not all women underwent more than one embryo transfer after an initial negative result, even when surplus frozen embryos were available. The only biologically plausible factor explaining the heterogeneity was the average number of embryo transfers performed per woman (Table 1). If the primary advantage of blastocyst transfer in fresh LBR is the autoselection of higher-quality embryos, it is reasonable to expect that cumulative live birth rate would converge between groups as the number of transfers increases. Consistent with this, two studies reported an average of 1.6 transfers per woman and found higher cumulative live birth rate with blastocyst transfer (Clua 2022; Ma 2024), whereas the study with an average of 2.2 transfers per woman found no such difference (Cornelisse 2024).

There may be little to no difference between the embryo stages in the miscarriage rate in fresh cycles (low-certainty evidence).

Women in the blastocyst-transfer groups may have fewer chances to freeze surplus embryos (low-certainty evidence). Cumulative live birth rate may be higher for blastocyst transfers in the initial transfers, but differences probably diminish as the number of cumulative transfers increases. If the reduced number of frozen embryos does not translate into a sustained improvement in cumulative success rates, the cost-effectiveness of blastocyst transfer may be lower, given the potential economic and emotional burden of undergoing more failed transfers; however, cost effectiveness was not assessed in the studies.

Blastocyst transfer may also increase the likelihood of having no embryos available for transfer (low-certainty evidence), largely due to developmental arrest before day 5. In many studies, this lower transfer rate resulted from a lack of viable blastocysts rather than deliberate policy. For some women, learning that embryos failed to develop by day 5 may be preferable to transferring cleavage-stage embryos with low potential for success; however, little research has explored the emotional impact of such outcomes [136]. Avoiding unnecessary embryo transfers must be weighed against the possible need for additional oocyte retrievals. Furthermore, extended culture could theoretically harm viable embryos under suboptimal laboratory conditions, and large variations in blastocyst development rates between trials highlight this concern. Indeed, we found some evidence that the process of extending culture to the blastocyst stage may negatively affect certain obstetric and perinatal outcomes.

Overall, we judged many of the included studies to be at high overall risk of bias, which reduced our confidence in the results, as reflected in the summary of findings table (Summary of findings 1). However, in most cases, sensitivity analyses excluding these studies showed similar results to our main analyses, supporting the robustness of the review findings.

Limitations of the evidence included in the review

The certainty of the evidence in this review was moderate for two outcomes—fresh live birth and cumulative preterm birth—and low for the others. The main reasons for downgrading were as follows.

- Risk of bias – blinding was not feasible for this intervention, and many studies had deviations from intended interventions, most often a higher number of embryos transferred in the cleavage-stage group than in the blastocyst group. In some cases, sequence generation and allocation concealment were inadequately reported.
- Inconsistency – we downgraded cumulative live birth rate for inconsistency (i.e. heterogeneity). While exploratory analyses suggested that the number of embryo transfers within follow-up might partly explain the heterogeneity, this remains a hypothesis and requires confirmation.

Other limitations of this review are the following.

- Restricted generalisability – most trials enroled women with good prognoses; only two small studies included women with poor prognoses. This limits the applicability of our findings to the broader IVF population.
- Incomplete outcome reporting – few studies reported cumulative live birth rates (3/36) or cumulative obstetric and perinatal outcomes (2/36), which are critical for assessing the overall benefit–risk profile.
- Variability in intervention details – differences in embryo culture media, freezing protocols, embryo transfer policies (Table 8), and use of adjunct technologies (e.g. time-lapse systems) may have influenced outcomes and contributed to between-study variability.
- Differences in the definition and reporting of cumulative live birth rate across studies represent a limitation. In particular, follow-up being limited to 12 months, and variation in whether all embryos were transferred may introduce heterogeneity in the estimates.

Overall, while the available evidence is sufficient to identify some important differences between blastocyst-stage and cleavage-stage transfers, gaps remain, particularly for poor-prognosis patients, cumulative outcomes, and safety endpoints, which should be addressed in future trials.

Limitations of the review processes

As far as possible, we adhered to the protocol methodology to limit any potential biases. However, we added new outcomes: cumulative pregnancy rates (derived from fresh and thawed cycles) in the 2016 update, and cumulative live birth rates, and cumulative obstetric and perinatal rates in the 2026 update. These outcomes reflect the policy of fresh embryo transfer and freezing remaining embryos for transfer later and more accurately reflect modern IVF practice than a single cycle transfer.

There is an important distinction between the outcomes 'live birth rate following fresh transfer' and ‘cumulative live birth rate’. In the last decade, many studies only reported the live birth rate following the first transfer, and trialists are generally used to reporting this outcome. Due to the extended embryo culture in blastocyst-stage transfers, the selected embryos are potentially of higher quality, but the number available for transfer is reduced, which could potentially affect the cumulative outcomes. Although live birth rates per fresh transfer are important and show benefits for blastocyst transfers, cumulative data suggest that these rates may become similar as the number of cumulative transfers increases. Additionally, we now have cumulative outcomes that could aid in the decision-making process: cumulative live birth rates are probably higher in the blastocyst group. In previous updates, these outcomes were absent, but this prior limitation has been mitigated by the inclusion of new studies.

The issue of publication bias is important in systematic reviews, as it may result in incorrect conclusions being reached. For example, the pressure for clinics to achieve high implantation rates with blastocyst culture could potentially lead to preferential publication of studies reporting favourable results. We explored this possibility using funnel plots for live birth and clinical pregnancy outcomes; these did not suggest the presence of publication bias. In addition, although six ongoing studies started more than five years ago have not yet published findings, which could raise concerns about publication bias, we did not downgrade the certainty of the evidence on this basis.

We also conducted sensitivity analyses excluding studies at high risk of bias, thereby including studies judged to be at low risk of bias or with some concerns. This approach was chosen to assess the robustness of the findings while retaining sufficient data for meaningful analysis. Sensitivity analyses restricted to studies at low risk of bias only were not undertaken, as the number of such studies was limited and these analyses were unlikely to be informative.

Another point to consider is the widely varying policies for assessing minimal quality of embryos for transfer that may have existed amongst trials. Some trials accepted transfer of developmentally-delayed embryos in blastocyst stage, whilst other trials were more selective and denied transfer of embryos that had not reached the late morula or early blastocyst stage. A strict selection policy for blastocyst transfer could have a potential bias in favour of the cleavage-stage arm for two outcomes: embryo freezing rate per woman and failure-to-transfer rate per woman. The freezing rate can be higher and the failure-to-transfer rate lower in the cleavage-stage group.

Blastocyst formation rates may also influence the pregnancy rate per embryo transfer for each trial, which ranged from 22.4% to 60% (Aziminekoo 2015; Schillaci 2002). Sequential media were used for the culture of blastocysts in the included trials that provided information on the media used. However, while most reported using various versions of Vitrolife G1/G2, others used a combination of different brands, or made the media in-house. This highlights the possibility that different brands and formulations influence the blastulation rates and subsequent outcomes. Improvements in culture media and embryo laboratory procedures should prompt new studies.

Finally, although we took steps to conduct an exhaustive search for eligible trials, it is possible that our search strategy failed to find some studies, resulting in bias.

Agreements and disagreements with other studies or reviews

This Cochrane review is in agreement with a systematic review published by Papanikolaou and colleagues in 2008, which reported that cycles where equal numbers of embryos were transferred had higher fresh live birth rates with blastocyst-stage transfers than with cleavage-stage transfers [137]. Our review also includes cumulative live birth (fresh and frozen-thawed cycles) rates, as well as cumulative obstetric and perinatal outcomes. Another systematic review was published by Wang and colleagues in 2014, which included only RCTs published after 2004, shows similar results with respect to live birth rates, but reported lower miscarriage rates in the blastocyst-stage group, which was not observed in our review [138]. Two published reviews that cited the previous version of this review are in agreement that better studies are required [139, 140], and argue that analysing obstetric and neonatal outcomes results is critical for drawing clearer conclusions. In agreement with those authors, we added those outcomes that were reported in only two studies. Another systematic review published by Martins and colleagues in 2017 emphasises that there is still too much uncertainty to conclude that transferring at one stage is better than the other [41].

Alviggi and colleagues published a systematic review of retrospective studies in 2018 that analysed perinatal outcomes [141]. They showed that blastocyst transfers may slightly increase the preterm and very preterm birth rates but may decrease the small-for-gestational-age rates in fresh cycles. However, these associations were not seen in frozen transfers. In 2020, Spangmose and colleagues published a large retrospective cohort study analysing obstetric and perinatal outcomes, after fresh blastocyst or cleavage transfer [49]. They showed that fresh blastocyst transfer increases the risk of placenta previa, preterm birth, and monozygotic twins. Furthermore, an altered male-female ratio with more males following blastocyst transfer was seen.

The same study group published a large retrospective cohort study [142], which analysed obstetric and perinatal outcomes, after frozen-thawed transfer of vitrified blastocysts or slow-frozen cleavage-stage embryos. They showed a higher risk of preterm birth after blastocyst-stage transfer. For maternal outcomes, no significant differences were found, although there was precision.

In agreement with them, we showed the increment of cumulative preterm birth in blastocyst transfers, but we did not find differences in the other outcomes.

Busnelli and colleagues published a systematic review including mostly retrospective studies that analysed the risk factors for monozygotic twins [143]. They showed that blastocyst transfer may be one of the factors that increases the risk of monozygotic twins.

AUTHORS' CONCLUSIONS

Implications for practice

Blastocyst-stage transfer may improve cumulative live birth rate (cLBR) within 12 months (including fresh and frozen-thawed cycles) more than cleavage-stage transfer, although this difference may diminish when additional frozen-thawed embryos derived from the same cycle are transferred.

There is moderate-certainty evidence that blastocyst-stage transfer probably increases live birth rates after fresh transfer, and low-certainty evidence that clinical pregnancy rates may also be higher. However, cumulative preterm birth rates are probably higher after blastocyst transfer, particularly for moderate/late preterm birth (moderate-certainty evidence).

Cleavage-stage transfer may lead to higher rates of embryo freezing and a lower likelihood of cycle cancellation due to no embryo being available for transfer (low-certainty evidence). There may be little to no difference between the embryo transfer stages in miscarriage rates for fresh transfers. Other obstetric and perinatal outcomes may differ between groups, but the evidence remains weak.

Most studies in the review recruited women with a good prognosis; therefore, findings should not be generalised to populations with a poor prognosis.

Choosing between blastocyst- and cleavage-stage transfer is complex, as it involves trade-offs across several outcomes. For those prioritising a shorter time to live birth with higher fresh live-birth rates, blastocyst-stage transfer may be preferable. However, patients and clinicians should also consider that blastocyst transfer may lead to a lower embryo freezing rate, higher risk of cycle cancellation, and possible increase in preterm birth. Given these competing outcomes and uncertainties, decisions should be individualised and made using shared decision-making tools and patient-centred counselling.

Implications for research

Although this review shows evidence of a difference in live birth rates from fresh embryos in favour of blastocyst-stage transfer compared to cleavage-stage transfer cycles, the findings for cumulative live birth rates (derived from fresh and thawed cycles) still raise questions. Future studies with longer follow-up must report live birth rates, cumulative live birth rates, miscarriage rates, and obstetric and perinatal outcomes, to enable consumers and service providers to make well-informed decisions on the best treatment option. Cumulative obstetric outcomes and time to pregnancy are also clinically meaningful endpoints that should be systematically evaluated in future studies. In addition, more studies evaluating women with a poor prognosis will provide more information about this important subgroup.

Based on the results of this review, we recommend the following, to ensure valuable data are produced.

- Adherence to Consolidated Standards of Reporting Trials (CONSORT) recommendations for randomised controlled trials (RCTs), especially methods to conceal participant assignment [144]
- Research into the best participant selection and inclusion criteria, considering prognostic factors such as age and outcomes from previous cycles
- Same media composition and brand for both groups up to the cleavage stage
- Explicit prespecified embryo transfer policies for both groups, preferably single-embryo transfer
- Long-term follow-up reports of cumulative live birth rates (including embryo thaws) presented as a survival analysis. A minimum of a one-year time period after randomisation for follow-up of cumulative live birth rates within studies (as part of study protocol), and a possible follow-up of results of transfers outside this time window in a later report (if applicable, i.e. if not all transfers have been performed within one year)
- Reporting of miscarriage, live birth, cumulative pregnancy and live birth rates, time to pregnancy, and reporting of cumulative obstetric and perinatal outcomes
- Given the post-hoc analysis that we performed by the number of embryo transfers, this prespecified subgroup analysis is warranted in RCTs.
- Given that the embryo freezing rate may be higher in cleavage-stage transfer cycles, more evidence about cost-effectiveness analysis is needed to draw firmer conclusions about value for money.
- An individual-patient data meta-analysis could help to identify specific subgroups of women who are more likely to benefit from blastocyst-stage embryo transfer compared with cleavage-stage transfer.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD002118.pub7.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

Supplementary material 9 Glossary

ADDITIONAL INFORMATION

Acknowledgements

Editorial and peer-reviewer contributions to the 2026 update of this review

Cochrane Gynaecology and Fertility supported the authors in the development of this Cochrane review update. Cochrane office in Auckland: Marian Showell for designing and running our database searches, and Helen Nagels for advising on responses to some aspects of the peer review feedback.

The following people conducted the editorial process for the 2026 version of the review.

- Sign-off Editor (final editorial decision): Madelon van Wely, Amsterdam UMC
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Joanne Duffield, Cochrane Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments, and supported the editorial team): Andrew Savage, Cochrane Editorial Service
- Copy Editor (copy editing and production): Laura MacDonald, Cochrane Central Production Service
- Peer-reviewers (provided comments and recommended an editorial decision): Wellington Martins, SEMEAR fertilidade, Reproductive Medicine, Brazil (clinical/content review); Professor Mohan S Kamath, Christian Medical College, Vellore, India (clinical/content review); Emma Axon, Cochrane Methods Support Unit (methods review); Jo Platt, Cochrane Editorial Information Specialist (search review)

Acknowledgments for previous versions of this review

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Dr Neil Johnson was a review author for the previous version of this review and made a significant contribution to the interpretation of results and performed some data extraction. David Olive, for the initial review, commented on drafts of the protocol and review. Michelle Proctor, for the initial review, was involved in selecting trials for inclusion, performed independent data extraction and risk of bias assessment, conducted the certainty of the evidence assessment of the included trials, and contributed to the discussion and interpretation of results. Quirine Lamberts, for the 2005 update, checked the data and study information extracted. Ariel Bardach, for the 2012 update, was involved in selecting trials for inclusion, performed independent data extraction, and performed the risk of bias assessment and certainty of the evidence assessment of the included trials.

The authors of the 2022 update acknowledge the contributions of Professor Cindy Farquhar to previous versions of this review. They also thank Dr Katie Stocking, Associate Professor Vanessa Jordan, Dr Selma Mourad, and Assistant Professor Sebastiaan Mastenbroek for providing peer review comments.

Contributions of authors

Demián Glujovsky: for the 2012, 2016, 2022 and 2026 updates: took the lead in writing the update of the review; performed new searches of databases for trials; involved in selecting trials for inclusion; performed independent data extraction and risk of bias and certainty of the evidence assessments of the included trials; was responsible for statistical analysis and interpretation of the data in the updates.

Simone Cornelisse: for the 2022 and 2026 updates: involved in study selection, data extraction, risk of bias and certainty of the evidence assessments, and final analysis and write-up of the text.

Deborah Blake: for the initial review and updates to 2005: took the lead in writing the protocol and review; performed initial searches of databases for trials; selected trials for inclusion; performed independent data extraction and assessed the risk of bias in the included trials; was responsible for statistical analysis and interpretation of the data. Also contributed to the final analysis and write up of the text of the 2012, 2016, 2022, and 2026 updates.

Andrea Quinteiro Retamar: for the 2016 and 2022 updates: selected trials for inclusion; performed independent data extraction and assessed the risk of bias of the included trials. In 2026 update, contributed to the final analysis and write-up of the text.

Cristian Alvarez Sedo: contributed to the final analysis and write-up of the text of the 2016 update. For the 2022 update: selected trials for inclusion; performed independent data extraction and assessed the risk of bias of the included trials. In 2026 update, contributed to the final analysis and write-up of the text.

Agustín Ciapponi: for the 2022 and 2026 updates: involved in study selection, risk of bias and certainty of the evidence assessments, and final analysis and write-up of the text.

Declarations of interest

Demián Glujovsky is the Scientific Director at a fertility clinic (Cegyr, part of the Eugin group) and undertakes private practice within those premises. He is an Editor with Cochrane Gynaecology and Fertility Group, but he was not involved in the editorial process for this review.

Simone Cornelisse is the lead author of an included study (Cornelisse 2024) and the lead author of an excluded study [145]. The included study was funded by a grant from the Leading the Change programme from Stichting Zorgevaluatie Nederland (project number 80-87600-98-19501), a Dutch foundation that consists of the Dutch Medical Specialists Federation, Dutch Health Insurers, and Dutch Patient Federation. Before receiving this grant, the Netherlands Organisation for Health Research and Development (ZonMw) externally peer-reviewed the study protocol. The funder of the study had no role in study design; data collection, analysis, or interpretation; or writing of the report. SC was not involved in decisions regarding study selection, data extraction and management, assessment of risk of bias or grading the certainty of the evidence. These tasks were performed by two review authors independently (DG, AC).

Deborah Blake is the Scientific Director of Repromed Auckland Ltd. She has no known conflicts of interest to declare.

Andrea Quinteiro Retamar is a medical doctor. She is the egg donor coordinator of a fertility clinic (Cegyr) and undertakes private practice within those premises. She is affiliated with the Sociedad Argentina de Medicina Reproductiva (SAMeR; Argentina Society of Reproductive Medicine).

Cristian Alvarez Sedo has no known conflicts of interest to declare.

Agustín Ciapponi works as a health professional at Servicio de Medicina Familiar y Comunitaria, Hospital Italiano de Buenos Aires, Argentina. He has no known conflicts of interest to declare.

Sources of support

Internal sources

- None, Other

  None

External sources

- None, Other

  None

Registration and protocol

Protocol (2000): protocol for the original Cochrane review comparing blastocyst-stage versus cleavage-stage embryo transfer in in vitro fertilisation; published internally by the Cochrane Gynaecology and Fertility Group.

Original review (2002) https://doi.org/10.1002/14651858.CD002118

Review update (2005) https://doi.org/10.1002/14651858.CD002118.pub2

Review update (2007) https://doi.org/10.1002/14651858.CD002118.pub3

Review update (2012) https://doi.org/10.1002/14651858.CD002118.pub4

Review update (2016) https://doi.org/10.1002/14651858.CD002118.pub5

Review update (2022) https://doi.org/10.1002/14651858.CD002118.pub6

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see Supplementary material 8: full search strategies for each database; full citations for each unique report for all studies (included, ongoing, or excluded at the full-text screening stage) in the final review; study data, including study information, study arms, and study results; consensus risk of bias assessments; and analysis data, including overall estimates, subgroup estimates, and individual data rows, with the corresponding analysis settings). Appropriate permissions have been obtained for the inclusion of this material. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from Excel are available from the authors upon reasonable request.

What's new

Date	Event	Description
23 January 2026	New citation required and conclusions have changed	• We included four new studies (one of which was awaiting classification in the last version of the review and another that was classified in the last version). • We included one new ongoing study. • Conclusions on cumulative rates have new data (we added the cumulative live birth rate). • Data on cumulative obstetric and perinatal outcomes were added. • We are no longer reporting cumulative pregnancy rates. • We changed the review title to correct the intervention order from 'Cleavage-stage versus blastocyst-stage embryo transfer in assisted reproductive technology' to 'Blastocyst-stage versus cleavage-stage embryo transfer in assisted reproductive technology'. • We changed the author order in the byline from 'Demián Glujovsky, Andrea Marta Quinteiro Retamar, Cristian Roberto Alvarez Sedo, Agustín Ciapponi, Simone Cornelisse, Deborah Blake' to 'Demián Glujovsky, Simone Cornelisse, Deborah Blake, Andrea Marta Quinteiro Retamar, Cristian Roberto Alvarez Sedo, Agustín Ciapponi'.
23 January 2026	New search has been performed	• An updated search was performed on 1 October 2024. • One critical outcome (cumulative live birth rate), one critical adverse outcome (cumulative preterm birth) and other important outcomes (other cumulative obstetric and perinatal outcomes) were incorporated. • Multiple pregnancy is no longer considered a critical outcome. • For methodological reasons, cumulative clinical pregnancy rate was excluded from the current report.

History

Protocol first published: Issue 2, 2000
Review first published: Issue 2, 2002

Date	Event	Description
21 June 2022	Amended	Correction of study data in abstract and summary of findings table
11 May 2020	New citation required but conclusions have not changed	The addition of five new RCTs has not led to a change in the conclusions of this review.
11 May 2020	New search has been performed	Updated with five new RCTs (Hatirnaz 2017; Kaser 2017; Levi-Setti 2018; Singh 2017; Yang 2018). Cindy Farquhar has been removed from the list of authors and Agustin Ciapponi and Simone Cornelisse have been added to the list of authors for this update.

Date	Event	Description
5 May 2016	New citation required and conclusions have changed	Updated May-June 2016
3 May 2016	New search has been performed	Updated with four new RCTs (Aziminekoo 2015; Fernandez-Shaw 2015; Gaafar 2015; Kaur 2014)
28 May 2013	Amended	Correction of author order
21 February 2012	New search has been performed	History of this review: the review was first published in *The Cochrane Library* in 2000 with 10 RCTs, and a journal paper version was published in Human Reproduction in 2003 with an additional four RCTs. The authors were Debbie Blake, Michelle Proctor, Neil Johnson and David Olive. There was no evidence for a difference in pregnancy rate and only one trial reported live birth rates. In the 2005 update, seven RCTs from the 2000 review were excluded (for quasi-randomisation or per cycle data only or other study design problems) and 13 new RCTs were added. In addition, the outcomes in Metaview were reconfigured. Cindy Farquhar and Quirine Lamberts assisted with the update and were added as authors. Some protocol changes were made to the outcome measures and to the sensitivity analysis (day of randomisation) and subgroup analyses (prognosis). More included trials reported live birth outcomes but there was still no evidence of a difference in success rates. The 2007 update had two new trials added, to bring the total to 18. There was a new subcategory for single embryo transfer. This update was performed by Debbie Blake, Neil Johnson and Cindy Farquar. It reported for the first time a significant difference in live birth and pregnancy outcomes in favour of blastocyst culture. In the 2012 update led by Demian Glujovsky with the assistance of Cindy Farquhar and Debbie Blake, five new studies were added (Brugnon 2010; Elgindy 2011; Fisch 2007; Pantos 2004; Ten 2011)
21 February 2012	New citation required but conclusions have not changed	5 new studies have been added. There are no changes to the conclusions reported in the 2007 update
29 August 2011	Amended	Plain Language Summary corrected
17 November 2010	Amended	New search strategies performed
11 November 2010	Amended	New author
23 July 2007	New citation required and conclusions have changed	Substantive amendment

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ADDITIONAL TABLES

Table 1. Important characteristics and variables from studies that reported cumulative live birth rate

Variable	Clua 2024	Cornelisse 2024	Ma 2024
N randomised women	134	1202	992
Age	26.2	33.9*	29.6
Inclusion criteria (minimum number of embryos)	No minimum (average 8.6 fertilised oocytes)	≥ 4 (on day 2)	≥ 3 (on day 3)
Follow-up for cumulative rates (months)	12	12	12
Retrieved oocytes	18.7 ± 9.7	11.5 ± 4.9	15.0 ± 7.9
Transferred embryos (D5 vs D3)	1.1 vs 1.4	1.1 vs 1.1	1.0 vs 1.0
Number of embryo transfers (D5 vs D3)	1.69 (1.51 vs 1.87)	2.18 (1.99 vs 2.37)	1.61 (1.46 vs 1.77)
Method of cryopreservation (D5 vs D3)	Vitrification	Vitrification: 84.7% vs 84.3% Slow freezing: 0.3% vs 8.5%	Vitrification
Fresh LBR (%) (D5 vs D3)	57.6 vs 33.8	37.0 vs 29.5	50.9 vs 40.4
Cumulative LBR (%) (D5 vs D3)	75.4 vs 59.4	58.9 vs 58.4	74.8 vs 66.3

D3: day 3 of embryo development (cleavage stage); D5: day 5 of embryo development (blastocyst stage); (c)LBR: (cumulative) live birth rate; N: number; NS: not significant; vs: versus
*When analysed only in women below 36 years old, cLBR had little difference (62.6% vs 67.1%, P > NS)

Table 2. Blastocyst formation and implantation rate (in day 5 to 6 transfers)

Study	Blastocyst formation rate	Implantation D2/3	Implantation D5/6	Other
Aziminekoo 2015	22.4%	21/173; 12.1%	22/152; 14.5%	-
Brugnon 2010	Not stated	24/52; 46.2%	23/55; 41.8%	-
Bungum 2003	55.2%	50/114; 43.9%	44/120; 36.7%	2/61 participants had only 1 blastocyst
Clua 2022	56.7%	34/97; 35.1%	44/72; 61.1%	-
Cornelisse 2024	56.2%	199/598 (fresh); 33.3% 388/1528; 25.4%(cumulative)	243/605 (fresh); 40.2% 378/1264; 29.9%(cumulative)	-

Table 2. Blastocyst formation and implantation rate (in day 5 to 6 transfers) *(Continued)*

Coskun 2000	28%	50/235; 21.3%	52/218; 23.9%	77% participants had at least 1 blastocyst
Devreker 2000	Not stated	1/34; 2.9%	8/19; 42.1%	-
Elgindy 2011	97%	71/197; 36%	53/280; 19%	-
Emiliani 2003	48%	57/197; 28.9%	50/168; 29.8%	-
Fernandez-Shaw 2015	67.7 %	20/71; 28.1%	36/84; 42.8%	-
Fisch 2007	Not stated	11/12; 92%	4/8; 50%	-
Frattarelli 2003	Not stated	18/69; 26.1%	23/53; 43.4%	-
Gaafar 2015	Not stated	Not stated	Not stated	-
Gardner 1998a	46.5%	64/174; 36.8%	53/95; 55.8%	85% women had at least 2 blastocysts
Hatirnaz 2017	52.6%	45/95; 47.4%	43/95; 45.3%	-
Hreinsson 2004	33%	29/139; 20.9%	24/114; 21.1%	2 morula replaced (one implanted). 60% pregnancy rate when top-quality blasts transferred
Karaki 2002	33%	37/291 12.7%	37/142; 26.1%	9/80 cancelled due to lack of blastocysts (unselected)
Kaser 2017	Not stated	23/56; 41.1%	26/53; 49.1%	-
Kaur 2014	Not stated	66/309; 21.4%	102/290; 35.2%	-
Kolibianakis 2004	50.7%	96/234; 41.0%	94/226; 41.6%	-
Levi-Setti 2018	Not stated	25.67%	28.37%	-
Levitas 2004	43%	4/56; 7.1%	10/24; 41.6%	Day 5 to 7 26% cancelled due to lack of blastocysts (poor prognosis)
Levron 2002	34.2%	53/137; 38.7%	20/99; 20.2%	6.5% cancelled due to lack of blastocysts (good prognosis)
Livingstone 2002	Not stated	-	-	-
Ma 2024	37.3%	Not stated	Not stated	-
Mahmoudinia 2024	NS	37/111; 33.33%	40/107; 37.38%	4.2% cancelled due to lack of blastocysts
Motta 1998	Not stated	51/262; 19.5%	36/120; 30.0%	6/58 cycles cancelled D5 no blastocysts
Pantos 2004	44.6%	15.8%	15.8%	-

Table 2. Blastocyst formation and implantation rate (in day 5 to 6 transfers) *(Continued)*

Papanikolaou 2005	Not stated	35/170; 20.6%	59/158; 37.3%	4/158 women had only 1 blast transferred due to lack of availability and 1 had it on request
Papanikolaou 2006	Not stated	38/156; 24%	58/149; 38.9%	Number of participants with no embryos available D3: 8 and D5: 11
Rienzi 2002	44.8%	34/96; 35.4%	38/100; 38.0%	Good prognosis
Schillaci 2002	60.3%	23/168; 13.7%	26/110; 23.6%	Unselected population nil cancellations D5
Singh 2017	Not stated	Not stated	Not stated	-
Ten 2011	Not stated	21/54; 38.9%	26/56; 46.4%	Good prognosis
Van der Auwera 2002	44.7%	31/106; 29.2%	41/90; 45.6%	27% cancellation D5 (unselected population)
Yang 2018	Not stated	80/290; 62.1%	22/306; 72.5%	-

D: day.
Morula=an early embryonic stage (around day 4 after fertilisation) consisting of a compact cluster of cells, preceding blastocyst formation.

Table 3. Overview of included studies and synthesis table illustrating key study characteristics (table 1 of 3)

Study (year)	Study design	Population (sample size: blastocyst/cleavage)	Outcomes with available data	**Notes**
Aziminekoo 2015	RCT	Poor prognosis (sample size: 57/61)	• Clinical pregnancy • Multiple pregnancy • Miscarriage • Failure-to-transfer	We contacted study authors for additional information; a reply was not received.
Brugnon 2010	RCT	Good prognosis (sample size: 52/56)	• Live birth • Clinical pregnancy	We contacted study authors for additional information; a reply was received clarifying the randomisation process, as well as information on live birth rates and the place where the study was done.
Bungum 2003	RCT	Good prognosis (sample size: 61/57)	• Clinical pregnancy • Multiple pregnancy • Miscarriage • Embryo freezing	We contacted study authors for additional information; a reply was received clarifying the randomisation process. Authors also confirmed that the included study reported on the same participants as another scientific abstract, which was consequently excluded.
Clua 2022	RCT	Good prognosis (sample size: 65/69)	• Cumulative live birth • Live birth • Clinical pregnancy	We contacted study authors for additional information; a reply was received confirming the follow-up time of outcome assessment, along with additional, unpublished data that were used in the analyses on cumulative pregnancy rates (analysis 3.1),

Table 3. Overview of included studies and synthesis table illustrating key study characteristics (table 1 of 3) *(Continued)*

			• Multiple pregnancy • Miscarriage	embryo freezing rate (analysis 8.1) and failure-to-transfer rate (analysis 9.1).
Cornelisse 2024	RCT	Good prognosis (sample size: 603/599)	• Cumulative live birth • Live birth • Clinical pregnancy • Miscarriage • Multiple pregnancy • Embryo freezing • Obstetric-perinatal outcomes	We contacted study authors for additional information; a reply was received clarifying differences in cryopreservation between the groups in terms of the freezing method used and the number of transferred embryos.
Coskun 2000	RCT	Good prognosis (sample size: 100/101)	• Clinical pregnancy • Multiple pregnancy • Miscarriage	-
Devreker 2000	RCT	Poor prognosis (sample size: 11/12)	• Live birth • Clinical pregnancy • Miscarriage	We contacted study authors for additional information regarding randomisation; we did not receive a reply.
Elgindy 2011	RCT	Not stated (sample size: 100/100)	• Live birth • Clinical pregnancy • Miscarriage • Failure-to-transfer	-
Emiliani 2003	RCT	Unselected (sample size: 82/89)	• Live birth • Clinical pregnancy • Multiple pregnancy • Failure-to-transfer	-
Fernandez-Shaw 2015	RCT	Unselected (sample size: 60/60)	• Cumulative live birth • Live birth • Clinical pregnancy • Multiple pregnancy • Miscarriage	We contacted study authors for additional information; a reply was received clarifying the randomisation and allocation process, protocol registration and live birth rates, along with additional, unpublished data, which were included in the analysis on live births (analysis 1.1).
Fisch 2007	RCT	Good prognosis (sample size: 12/8)	• Clinical pregnancy	We contacted study authors for updated information; a reply was received but no updated data were added.
Frattarelli 2003	RCT	Good prognosis (sample size: 26/23)	• Live birth • Clinical pregnancy	We contacted study authors for additional information; a reply was received clarifying the randomisation process.

Table 3. Overview of included studies and synthesis table illustrating key study characteristics (table 1 of 3) *(Continued)*

			• Multiple pregnancy • Miscarriage • Failure-to-transfer	

RCT: randomised controlled trial; vs: versus

Table 4. Overview of included studies and synthesis table illustrating key study characteristics (table 2 of 3)

Study (year)	Study design	Population (sample size: blastocyst/cleavage)	Outcomes with available data	**Notes**
Gaafar 2015	RCT	Unselected (sample size: 126/126)	• Clinical pregnancy • Miscarriage	-
Gardner 1998a	RCT	Good prognosis (sample size: 45/47)	• Clinical pregnancy • Embryo freezing	-
Hatirnaz 2017	RCT	Unselected (sample size: 9595)	• Live birth • Clinical pregnancy • Miscarriage • Multiple pregnancy	-
Hreinsson 2004	RCT	Good prognosis (sample size: 64/80)	• Clinical pregnancy	We contacted study authors for additional information; a reply was received clarifying the randomisation process.
Karaki 2002	RCT	Unselected (sample size: 80/82)	• Clinical pregnancy • Multiple pregnancy • Miscarriage • Failure-to-transfer	We contacted study authors for additional information; a reply was received clarifying the randomisation process.
Kaser 2017	RCT	Good prognosis (sample size: 107/56)	• Clinical pregnancy	-
Kaur 2014	RCT	Good prognosis (sample size: 150/150)	• Clinical pregnancy • Multiple pregnancy • Miscarriage	-
Kolibianakis 2004	RCT	Unselected (sample size: 226/234)	• Clinical pregnancy • Miscarriage • Embryo freezing • Failure-to-transfer	-

Table 4. Overview of included studies and synthesis table illustrating key study characteristics (table 2 of 3) *(Continued)*

Levi-Setti 2018	RCT	Good prognosis (sample size: 194/194)	• Live birth • Clinical pregnancy • Multiple pregnancy • Miscarriage	-
Levitas 2004	RCT	Poor prognosis (sample size: 23/31)	• Live birth • Clinical pregnancy • Multiple pregnancy • Miscarriage • Failure-to-transfer	We contacted study authors for additional information; a reply was received clarifying the randomisation process.
Levron 2002	RCT	Good prognosis (sample size: 46/44)	• Live birth • Clinical pregnancy • Multiple pregnancy • Embryo freezing • Failure-to-transfer	We contacted study authors for additional information; a reply was received clarifying the randomisation process.
Livingstone 2002	RCT	Good prognosis (sample size: 30/29)	• Live birth • Clinical pregnancy • Multiple pregnancy • Miscarriage	-

RCT: randomised controlled trial; vs: versus

Table 5. Overview of included studies and synthesis table illustrating key study characteristics (table 3 of 3)

Study (year)	Study design	Population (sample size: blastocyst/cleavage)	Outcomes with available data	**Notes**
Ma 2024	RCT	Good prognosis (sample size: 497/495)	• Cumulative live birth • Live birth • Clinical pregnancy • Miscarriage • Embryo freezing • Obstetric-perinatal outcomes	-
Mahmoudinia 2024	RCT	Good prognosis (sample size: 120/120)	• Live birth rate • Clinical pregnancy • Miscarriage	-

Table 5. Overview of included studies and synthesis table illustrating key study characteristics (table 3 of 3) *(Continued)*

			• Multiple pregnancy	
Motta 1998	RCT	Unselected (sample size: 58/58)	• Clinical pregnancy • Multiple pregnancy • Embryo freezing • Failure to transfer	-
Pantos 2004	RCT	Unselected (sample size: 81/162)	• Clinical pregnancy • Embryo freezing	-
Papanikolaou 2005	RCT	Good prognosis (sample size: 80/84)	• Live birth • Clinical pregnancy • Miscarriage • Multiple pregnancy • Failure to transfer	We contacted study authors for additional information; a reply was received regarding concealment.
Papanikolaou 2006	RCT	Good prognosis (sample size: 176/175)	• Live birth • Clinical pregnancy • Multiple pregnancy • Miscarriage • Embryo freezing • Failure to transfer	We contacted study authors for additional information; a reply was received regarding concealment, media, and freezing.
Rienzi 2002	RCT	Good prognosis (sample size: 50/48)	• Live birth • Clinical pregnancy • Miscarriage • Multiple pregnancy • Embryo freezing • Failure to transfer	We contacted study authors for additional information; a reply was received clarifying the randomisation process.
Schillaci 2002	RCT	Unselected (sample size: 60/60)	• Clinical pregnancy	-
Singh 2017	RCT	Good prognosis (sample size: 243/195)	• Clinical pregnancy	-
Ten 2011	RCT	Good prognosis (sample size: 28/27)	• Clinical pregnancy • Embryo freezing	-
Van der Auwera 2002	RCT	Unselected (sample size: 70/66)	• Live birth • Clinical pregnancy • Multiple pregnancy	-

Table 5. Overview of included studies and synthesis table illustrating key study characteristics (table 3 of 3) *(Continued)*

			• Miscarriage • Embryo freezing • Failure to transfer	
Yang 2018	RCT	Good prognosis (sample size: 300/300)	• Clinical pregnancy • Multiple pregnancy • Miscarriage	-

RCT: randomised controlled trial; vs: versus

Table 6. Culture techniques of included studies

Trial	Culture technique day 2/3	Culture technique day 5/6
Aziminekoo 2015	Sydney IVF cleavage medium, Cook	Sydney IVF blastocyst medium
Brugnon 2010	G series medium (Vitrolife, Sweden)	G series medium (Vitrolife, Sweden)
Bungum 2003	Sequential G1 Vitrolife	Sequential G1/G2 Vitrolife
Clua 2022	G series medium (Vitrolife, Sweden)	G series medium (Vitrolife, Sweden)
Cornelisse 2024	NS	NS
Coskun 2000	Sequential MediCult	Sequential G1/G2 Vitrolife
Devreker 2000	NS	NS
Elgindy 2011	NS	NS
Emiliani 2003	In-house sequential (based on G1/G2)	In-house sequential (based on G1/G2)
Fernandez-Shaw 2015	Sequential G1 Vitrolife	Sequential G1/G2 Vitrolife
Fisch 2007	NS	NS
Frattarelli 2003	NS	NS
Gaafar 2015	NS	NS
Gardner 1998a	Single Ham's F10 In-house	Sequential G1/G2 In-house
Hatirnaz 2017	Standard culture medium	Sequential G1/G2 Scandinavian IVF Sciences
Hreinsson 2004	Vitrolife IVF	Sequential G1/G2 or CCM Vitrolife
Karaki 2002	MediCult	Sequential G1/G2 Vitrolife
Kaser 2017	Global total with HSA LifeGlobal	Global total with HSA LifeGlobal
Kaur 2014	Cleavage medium	G2 Plus media
Kolibianakis 2004	Sequential G1 Vitrolife	Sequential G1/G2 Vitrolife

Table 6. Culture techniques of included studies *(Continued)*

Levi-Setti 2018	NS	NS
Levitas 2004	NS	Sequential - G1/G2 Vitrolife
Levron 2002	NS	NS
Livingstone 2002	Sequential - Sydney IVF Cook	Sequential - Sydney IVF Cook
Ma 2024	NS	Sequential
Mahmoudinia 2024	Sequential	Sequential
Motta 1998	Sequential - Irvines P1	Sequential - Irvines P1 then Blast media
Pantos 2004	NS	NS
Papanikolaou 2005	Sequential - Vitrolife G1/G2 GII or GIII	Sequential - Vitrolife G1/G2 GII or GIII
Papanikolaou 2006	Assume sequential - Vitrolife G1/G2	Assume sequential - Vitrolife G1/G2
Rienzi 2002	Sequential G1 Vitrolife	Sequential G1/G2 Vitrolife
Schillaci 2002	NS	NS
Singh 2017	NS	NS
Ten 2011	NS	NS
Van der Auwera 2002	Sequential both Cook and Vitrolife	Sequential both Cook and Vitrolife
Yang 2018	Sequential media (G1.5, Vitrolife) in a Primo Vision time-lapse system (Vitrolife)	Sequential G1/G2 Vitrolife

CCM: a Vitrolife trademarked medium for blastocyst culture
IVF: in vitro fertilisation
G1/G2: sequential media from Vitrolife
NS: not stated

Table 7. Comparison of the two studies assessing cumulative obstetric and perinatal outcomes

Outcome	Cornelisse 2024		Ma 2024	
	Blastocyst	Cleavage	Blastocyst	Cleavage
Stillbirth or intrauterine death	0/603 (0%)	1/599 (0.2%)	8/497 (1.6%)	4/495 (0.8%)
Neonatal death	0/603 (0%)	0/599 (0%)	1/497 (0.2%)	2/495 (0.4%)
Major congenital anomalies	4/603 (0.7%)	4/599 (0.7%)	11/497 (2.2%)	9/495 (1.8%)
Preterm birth (< 37 weeks)	34/603 (5.6%)	20/599 (3.3%)	30/497 (6.0%)	18/495 (3.6%)
Early preterm birth (< 32 weeks)	3/603 (0.5%)	4/599 (0.7%)	5/497 (1.0%)	4/495 (0.8%)
Small for gestational age (< 10 centile)	28/603 (4.6%)	31/599 (5.2%)	23/497 (4.6%)	20/495 (4.0%)

Table 7. Comparison of the two studies assessing cumulative obstetric and perinatal outcomes *(Continued)*

Large for gestational age (> 90 centile)	28/603 (4.6%)	30/599 (5.0%)	63/497 (12.7%)	54/495 (10.9%)
Gestational diabetes mellitus	-	-	54/497 (10.8%)	55/495 (11.1%)
Placenta previa	-	-	8/497 (1.6%)	2/495 (0.4%)
Placental abruption	-	-	3/497 (0.6%)	2/495 (0.4%)
Placental accreta	-	-	12/497 (2.4%)	6/495 (1.2%)
Postpartum haemorrhage	-	-	10/497 (2.0%)	17/495 (3.4%)

Table 8. Mean number of embryos transferred

Study ID	Day 2/3	Day 5/6
Aziminekoo 2015	2.8 ± 1.1	2.6 ± 0.6
Brugnon 2010	1.0 ± 0	1.0 ± 0
Bungum 2003	2.0 ± NS	2.0 ± NS
Clua 2022	1.4 ± 0.5	1.1 ± 0.5
Cornelisse 2024	1.1 ± NS	1.1 ± NS
Coskun 2000	2.3 ± 0.6	2.2 ± 0.5
Devreker 2000	2.8 ± NS	1.7 ± NS
Elgindy 2011	2.8 ± 0.4	2.0 ± 0.2
Emiliani 2003	2.1 ± 0.4	1.9 ± 0.3
Fernandez-Shaw 2015	1.5 ± 0.5	1.4 ± 0.5
Fisch 2007	1.0 ± 0	1.0 ± 0
Frattarelli 2003	3.0 ± 0.5	2.0 ± 0.2
Gaafar 2015	NS	NS
Gardner 1998a	3.7 ± 0.1	2.2 ± 0.1
Hatirnaz 2017	1.4 ± 0.6	1.4 ± 0.4
Hreinsson 2004	1.8 ± NS	1.9 ± NS
Karaki 2002	3.5 ± 0.6	2.0 ± 0.1
Kaser 2017	1.0 ± 0	1.0 ± 0
Kaur 2014	2.0 ± 0.7	1.9 ± 0.5

Table 8. Mean number of embryos transferred *(Continued)*

Kolibianakis 2004	1.9 ± 0.1	1.8 ± 0.1
Levi-Setti 2018	1.9 ± 0.4	1.8 ± 0.6
Levitas 2004	3.4 ± NS	1.9 ± NS
Levron 2002	3.1 ± 0.6	2.3 ± 0.8
Livingstone 2002	2.0 ± NS	1.0 ± NS
Ma 2024	1.0 ± NS	1.1 ± NS
Mahmoudinia 2024	NS*	NS*
Motta 1998	4.6 ± NS	2.3 ± NS
Pantos 2004	4.0 ± 1.5	3.4 ± 1.1
Papanikolaou 2005	2.0 ± 0	2.0 ± 0.5
Papanikolaou 2006	1.0 ± 0	1.0 ± 0
Rienzi 2002	2.0 ± 0	2.0 ± 0
Schillaci 2002	2.8	1.8
Singh 2017	NS	NS
Ten 2011	2.0 ± NS	2.0 ± NS
Van der Auwera 2002	1.9 ± 0.3	1.9 ± 0.2
Yang 2018	1.0 ± 0	1.0 ± 0

NS: not stated
*"Considerably more cleavage-stage embryos were transferred compared to the number of transferred blastocysts"